

CHAPTER 12

TESTING STATISTICAL HYPOTHESES

12.1 INTRODUCTION

In this chapter, we present the basic concepts and results of the theory of testing statistical hypotheses.

Assume that a random sample $X_1, X_2, \dots, X_n$ was selected from distribution $f(x, \theta)$, where θ is an element of a parameter space Θ . Unlike in the estimation problems, we are now not interested in approximating the true value of θ . What we need to know is only whether θ belongs to some specific subset $\Theta_0 \subset \Theta$. A procedure that allows us to decide whether or not θ is in $\Theta_0 \subset \Theta$ is called a *test of the hypothesis* that $\theta \in \Theta_0$.

The problem is trivial in the case where the sets of possible values of X are disjoint for $\theta \in \Theta_0$ and $\theta \notin \Theta_0$ ($\theta \in \Theta_1$). However, in the more realistic cases, some (or even all) values of $\mathbf{X}$ can occur for both $\theta \in \Theta_0$ and for $\theta \notin \Theta_0$, and there is no procedure that will *always* allow us to reach the correct decision. Indeed, it may happen that while θ is in Θ_0 , we will observe a value of $\mathbf{X}$ that will lead us to the conclusion that $\theta \notin \Theta_0$. On the other hand, it may happen that θ is not in Θ_0 , and we will reach the conclusion that $\theta \in \Theta_0$. These two possibilities are called errors of type I and type II, and the objective is to choose the decision rule that will minimize their probabilities.

■ EXAMPLE 12.1

A consumer protection agency decides to investigate complaints that some boxes contain less of a product (e.g., cereal) than the amount printed on the box indicates. The boxes are filled and then sealed by a machine. Even with the most sophisticated equipment available, the weight of the cereal put into boxes will vary. Suppose that it is established that these amounts follow a normal distribution with some mean μ (which can be set in the packing plant) and a standard deviation σ (which is the attribute of the machine). For example, let $\mu = 20$ oz and $\sigma = 1.5$ oz.

If the mean μ is set at 20 oz, then about 50% of the boxes will contain less than the nominal amount, and at least for buyers of these boxes, the fact that the other 50% of boxes contain more than the nominal amount may be of no relevance. Consequently, the packing company must set the average μ at a value *above* the nominal weight 20 oz. Since no value μ will guarantee that the content of a box will *always* be above 20 oz, a reasonable requirement may be that 99% of boxes must contain at least 20 oz of cereal. If X denotes the weight of the cereal in a randomly chosen box, then

$$0.99 = P(X \geq 20) = P\left(Z \geq \frac{20 - \mu}{1.5}\right) = 1 - \Phi\left(\frac{20 - \mu}{1.5}\right),$$

and $\mu = 23.5$ oz. Thus, in order to satisfy the requirement, the company should set the average weight of the content of the box at least at a value 23.5 oz.

The agency that investigates the customers' complaints has to determine whether or not the average weight content is at least 23.5 oz. Since opening all boxes and weighting their content is not feasible, the agency must decide how to do it in a more practical way. For example, one may agree (and such a decision should be made in advance, prior to any sampling) that $n = 16$ boxes of the cereal in question will be bought; the stores will be sampled, and in each store, one box will be chosen at random from the shelf. Then, the content of all 16 boxes will be weighed and the average $\bar{x}$ will be calculated. Next some threshold has to be established (e.g., $\bar{x} = 23$) so that if $\bar{x} \leq 23$ oz, the hypothesis $\mu \geq 23.5$ will be rejected; otherwise, it will be accepted.

The probabilities of two types of errors can be calculated as follows: It may happen that $\mu \geq 23.5$, but the sample average $\bar{x}$ falls below 23 oz. We then declare the company to be at fault (of *not* putting, on average, at least 23.5 oz of cereal in boxes), although in fact the company is not at fault.

It may also happen that we observe $\bar{x} > 23$ (hence we declare the company as not being at fault), whereas in fact $\mu < 23.5$. To compute the chance of these two kinds of errors, we will find the probability that the company is declared at fault while the true mean is μ ,

$$P\{\bar{X} \leq 23 | \mu\} = \Phi\left(\frac{23 - \mu}{1.5/\sqrt{16}}\right),$$

which is a decreasing function of μ .

Declaring the company at fault is an error if in fact $\mu \geq 23.5$ (otherwise, it is a correct decision). The probability of such an error attains the smallest value for $\mu = 23.5$:

$$\Phi\left(\frac{23 - 23.5}{1.5/\sqrt{16}}\right) = 0.09.$$

The chances of the other type of error—declaring the company as not at fault, when in fact $\mu < 23.5$ —for any specific μ are equal to

$$P\{\bar{X} > 23 | \mu\} = 1 - \Phi\left(\frac{23 - \mu}{1.5/\sqrt{16}}\right).$$

For instance, if $\mu = 22$, then the probability above is less than 0.004.

To make a decision “ $\theta \in \Theta_0$ ” or “ $\theta \notin \Theta_0$,” we specify a set C of points in the space of values of $\mathbf{X}$ with the instruction: if the actual observation $\mathbf{x}$ is in C , then decide that $\theta \notin \Theta_0$;

otherwise, decide that $\theta \in \Theta_0$. Such a set C is called a *critical* (or *rejection*) *region* for the hypothesis that $\theta \in \Theta_0$.

In Example 12.1 we have $\mathbf{X} = (X_1, \dots, X_{16})$, and the critical region C consisting of all points $(x_1, \dots, x_{16})$ is such that $(1/16)(x_1 + \dots + x_{16}) \leq 23$.

Ideally, the best critical region C would minimize the probabilities of errors of type I and type II, but this principle is too general. One has to realize the following two points:

1. If $\theta \in \Theta_0$, then deciding that $\theta \notin \Theta_0$ is the error of type I. This error occurs whenever the observation $\mathbf{X}$ falls into the critical region C . So the probability of an error of type I is

$$\alpha(\theta) = P_\theta\{\mathbf{X} \in C\}. \quad (12.1)$$

On the other hand, if $\theta \notin \Theta_0$, then deciding that $\theta \in \Theta_0$ is the error of type II; this occurs whenever $\mathbf{X} \notin C$. So the probability of an error of type II is

$$\beta(\theta) = P_\theta\{\mathbf{X} \notin C\} = 1 - P_\theta\{\mathbf{X} \in C\}. \quad (12.2)$$

Probabilities of errors are not single numbers but functions of the parameter θ , both expressible in terms of $\pi(\theta) = P_\theta\{\mathbf{X} \in C\}$, called the *power function* of the test C . The minimization principle requires that power function be “as small as possible” for $\theta \in \Theta_0$ and “as large as possible” for $\theta \notin \Theta_0$. Obviously, the best test is such that $P_\theta\{\mathbf{X} \in C\} = 0$ if $\theta \in \Theta_0$ and $P_\theta\{\mathbf{X} \in C\} = 1$ if $\theta \notin \Theta_0$. However, in most cases of practical importance and interest, $P_\theta\{\mathbf{X} \in C\}$ is a continuous function of θ , and such a “best” test does not exist.

2. Looking at formulas (12.1) and (12.2), we see that any modification of C has opposite effects on the probabilities of errors of type I and type II: if one of them decreases, then the other increases.

Example 12.2, which is a continuation of Example 12.1, provides more explanation.

■ EXAMPLE 12.2

The properties of the decision procedure as to whether the cereal-producing company should be declared at fault depend on the choice of the threshold (23 oz), and the sample size n . Changing the threshold up or down changes the probabilities of the two types of errors, with the changes always going in opposite directions. The only way to decrease probabilities of both types of errors at the same time is to increase the sample size n . The “best” test must result from a compromise between the cost of sampling, and the consequences of the two types of errors. In the case of boxes of cereal, the cost of increasing n from 16 to 100 is of a little concern. Of course, increasing the sample size in other experiments might be more expensive, time-consuming, or even impossible.

The consequences of declaring the company at fault, when it is not, can again be expressed by the cost (the company may have to pay a fine, unnecessarily reset the mean to a higher value, etc.). The consequences of declaring company not at fault, when in fact it is, are hard to express in terms of cost; they involve many very small losses suffered by individual buyers.

The fact that the probabilities of errors of types I and II are negatively related, and that each is a function of θ (defined on sets Θ_0 and Θ_1 , respectively), makes formulation of the criterion to be optimized a difficult and challenging task. The conceptual structure of the theory is as follows. First, the problem of optimization is solved for the simplest case, where $\Theta_0 = \{\theta_0\}$ and $\Theta_1 = \{\theta_1\}$. The solution is given by the Neyman–Pearson lemma, which

determines the test (i.e., critical region) with the preassigned probability of error of type I, and a minimal probability of error of type II (or equivalently, a maximum power at value θ_1). Such test is called *most powerful*. In some sense, the Neyman–Pearson lemma plays a role analogous to that of the Cramér–Rao inequality in estimation theory: both set the standard by showing how much can potentially be achieved.

In some classes of testing problems, in which Θ_1 (and often also Θ_0) consist of more than one element, there exists a test C that is the most powerful against any $\theta \in \Theta_1$; such tests are called *uniformly most powerful* (UMP).

There are, however, classes of testing problems in which the UMP test does not exist. In such cases, the idea is to eliminate tests with some undesirable properties. The UMP test will often exist in the reduced class.

We discuss only one such reduction, reduction to the class of unbiased tests. The uniformly most powerful unbiased (UMPU) tests may not achieve maximum possible power specified by the Neyman–Pearson lemma, but they are uniformly most powerful among the unbiased tests (UMPU).

A following analogy can help one understand why such a reduction can lead to the existence of a UMP test. Suppose that we want to find the strongest athlete, in the sense of the strength of his right arm and strength of his left arm (considered separately). It may happen that there will be no strongest athlete in the group. The one with strongest right arm will have weak left arm, and vice versa, and no winner will be found. But one may restrict the competition to only those athletes that have equal strength in both arms. Among those, the winner (or winners) can always be found. Such a winner may be called the most powerful “unbiased” athlete.

This analogy immediately suggests another solution: Why not compare the athletes with respect to combined strengths of right and left arms? Or, according to the sum of appropriate scores, possibly different for left and right arms? (similar to deciding the winner in events such as the decathlon, which is based on sum of scores for different events).

While in our example the “most powerful unbiased athlete(s)” could always be found, it is not necessarily so with UMP tests. To understand why, suppose that the competition involves comparisons of strengths of right and left arms, and also right and left legs (separately). If no “absolute” winner (strongest on each limb) exists, one could reduce the competition to “unbiased” athletes, those whose strengths of right and left arms are the same, and whose strengths of right and left legs are the same. But now there is no guarantee that “most powerful unbiased athlete” exists, since an athlete with strongest arms need not have strongest legs, and vice versa.

In short, UMPU tests exist in some classes of problems and do not exist in other classes, so some further reduction may be needed to single out the UMP tests in this reduced class. The best exposition of the theory can be found in the monograph of Lehmann and Romano (2005).

The following example shows how reality can force one to use tests that are suboptimal:

■ EXAMPLE 12.3

The testing procedure suggested in Example 12.1 declares the company at fault when $\bar{x} < 23$. Such a test may well be optimal from the point of view of statistical criteria involving probabilities of errors, but it has a disturbing feature: it may happen that we declare the company at fault when all boxes tested actually contain more than the nominal 20 oz amount. Such a decision may not be defensible in court, in case the company decides to appeal. One should then restrict the consideration to procedures, that have the threshold value set below the nominal value 20 oz, and such procedures may be suboptimal.

12.2 INTUITIVE BACKGROUND

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution that belongs to some family $f(x, \theta)$ and we know that θ is an element of a parameter space Θ . As before, we do not know what is the true value of θ that governs the probabilities of the sample $X_1, X_2, \dots, X_n$.

The difference between the problem of estimation and the problem of testing is that now we are not interested in approximating the true value of θ . What we need to find out is only whether θ belongs to some specific subset $\Theta_0 \subset \Theta$.

■ EXAMPLE 12.4

Consider a politician who will win or lose the election, depending whether the proportion of voters who will cast their votes for him exceeds 50% or not. If θ is the fraction of voters who favor the politician in question, then $0 \leq \theta \leq 1$, so that the parameter space is $\Theta = [0, 1]$, and the set of interest is $\Theta_0 = (\frac{1}{2}, 1]$. A survey may give a sample $X_1, \dots, X_n$ of random variables with Bernoulli distribution $f(x, \theta) = \theta^x(1 - \theta)^{1-x}$, $x = 0, 1$, and the question is how to use the observation to determine whether or not $\theta \in \Theta_0$. A typical question may be: suppose that out of $n = 400$ voters sampled, only 195 will vote for the politician in question, and 205 will vote against him. Should one reject the hypothesis that $\theta > 1/2$? The answer depends on many factors, the most crucial being: how likely is it to observe the data so much (195 vs. 205) or even more in favor of rejecting the claim $\theta > 1/2$, if in fact the true value of θ is above $1/2$?

In the generally accepted terminology, the statement that θ belongs to Θ_0 and the statement that θ does not belong to Θ_0 are called *hypotheses*.

Let $\Theta_1 = \Theta \setminus \Theta_0$ so that out of the two hypotheses, $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \in \Theta_1$, exactly one must be true. Logically speaking, denying the truth of one hypothesis is equivalent to accepting the truth of the other. The actually observed sample $\mathbf{x} = (x_1, x_2, \dots, x_n)$ provides some evidence, typically pointing in favor of one of the two hypotheses. A *decision rule* is a rule that tells us what to do in case of observing $\mathbf{x} = (x_1, \dots, x_n)$. These actions may be “accept H_0 ,” “reject H_0 (and therefore accept H_1),” “take another observation,” or even “leave the matter unresolved.” A decision rule will be called a *test*²⁵ (more precisely, a *nonrandomized* test) if the only actions allowed are “accept H_0 ” and “accept H_1 .”

Since only two actions are allowed in a test, we must choose one of them regardless of the value of $\mathbf{x}$. Therefore, any decision rule is equivalent to a partition of the sample space into two subsets: the set of those points $\mathbf{x}$ at which H_0 is to be accepted, and its complement, the set of those points $\mathbf{x}$ at which H_0 is to be rejected (i.e., H_1 accepted).

Accordingly, we designate one of the hypotheses as the *null hypothesis*, denoted H_0 , with the corresponding subset of parameter space denoted Θ_0 . The complementary hypothesis is called the *alternative hypothesis*, and is usually denoted by H_1 or H_a . The corresponding subset of parameter space is $\Theta_1 = \Theta \setminus \Theta_0$.

The testing procedure is specified by the subset C of the sample space, called the *critical* region, and the rule

$$\text{reject } H_0 \text{ if } \mathbf{x} \in C.$$

²⁵ It is important to realize that the phrases “accept H ” and “reject H ” are to be interpreted as decisions regarding future behavior rather than regarding truth or falsehood of H . Thus, to “accept H ” means (in majority of practical situations) to proceed “as if H were true.” In fact it would be unrealistic to expect more: our decision is based on observation of a random sample; hence, we are always exposed to a risk of a wrong decision, namely rejecting H when in reality H is true, or accepting H when in reality it is false. The best one can do is to control the probabilities of these errors. Instead of stating that “hypothesis H_0 is accepted,” the verdict is often phrased in a more cautious way such as “there is not enough evidence to reject H_0 .”

Observe that in this formulation the two hypotheses, H_0 and H_1 , are treated symmetrically, and the designation of one of them as the null hypothesis is arbitrary. Subsequently, this symmetry will be lost, and the null hypothesis will play a different role than the alternative.

To formulate the meaningful criteria that will eventually allow to choose the best test, we introduce some auxiliary definitions.

Definition 12.2.1 A hypothesis $H: \theta \in \Theta_0$ is called *simple* if it completely specifies the distribution of the sample; otherwise, it is called *composite*. $\square$

As mentioned before, two types of error can be made as a result of our decision: a true null hypothesis can be rejected, or a false null hypothesis can be not rejected.

It goes without saying that we would like to minimize the probabilities of both types of errors. The trouble here is twofold. First, the probability of errors depends typically on the value of θ , so the phrase “probability of rejecting null hypothesis when it is true” does not have unique meaning if the null hypothesis is composite. In fact, there are many values of θ for which null hypothesis is true, and the probability in question depends on the particular value of θ . The same remark applies to the alternative hypothesis if it is composite.

The second source of difficulty lies in the fact that under any reasonable definition of the probability of errors, the two probabilities are inversely related. If the probability of one type of error decreases, then the probability of error of the other type increases. To obtain a convenient tool for expressing the probabilities of errors associated with a test, we introduce the following definition, central for the whole theory of testing statistical hypotheses:

Definition 12.2.2 If C is the critical region of a test of the null hypothesis $H_0: \theta \in \Theta_0$, the function $\pi_C(\theta)$, defined on parameter space Θ as

$$\pi_C(\theta) = P_\theta\{\mathbf{X} \in C\}, \quad (12.3)$$

is called the *power* (or *power function*) of test C . $\square$

Thus, power is the probability of rejecting the null hypothesis if the value of the parameter is θ . Qualitatively speaking, a good test should have high power when $\theta \notin \Theta_0$, since for such θ the null hypothesis is not true, and rejecting it is a correct decision. On the other hand, for $\theta \in \Theta_0$ the power of a good test should be low, since for such θ the null hypothesis is true, and rejecting it is a wrong decision. Let us now consider a few examples.

■ EXAMPLE 12.5

A supermarket buys oranges from a certain company, that claims that the fraction of unacceptable fruit (e.g., rotten) in each shipment does not exceed 3%. The supermarket is willing to accept such (but not higher) percentage of bad fruit. The procedure agreed upon specifies that a random sample of 30 oranges will be selected from each shipment (the method of sampling is typically also a part of the protocol). The shipment will be accepted if there is no more than one unacceptable fruit in this sample, and rejected (or bought at a discounted price, etc.) otherwise.

First, assume that the results of sampling constitute independent Bernoulli trials with a probability of “success” (finding a rotten orange) equal to θ .

The range of θ is (at least theoretically) the whole interval $\Theta = [0, 1]$. The random sample $X_1, \dots, X_{30}$ is drawn from the distribution $f(x, \theta) = \theta^x(1 - \theta)^{1-x}$, $x = 0, 1$. The supplier’s claim is that $\theta \leq 0.03$. Let us take this claim as the null hypothesis so that $\Theta_0 = [0, 0.03]$, while the alternative is $\Theta_1 = (0.03, 1]$.

The agreed procedure specifies that the supplier's claim (null hypothesis) is rejected if there are two or more bad oranges in the sample, that is, if

$$S_{30} = \sum_{i=1}^{30} X_i \geq 2.$$

This means that the critical region is the set $C = \{2, 3, \dots, 30\}$. The elements in C are values of S_{30} that lead to rejection of the supplier's claim that $\theta \in \Theta_0$.

The power function of this test is

$$\begin{aligned} \pi_C(\theta) &= P_\theta\{S_{30} \geq 2\} = 1 - P_\theta\{S_{30} = 0\} - P_\theta\{S_{30} = 1\} \\ &= 1 - (1 - \theta)^{30} - 30\theta(1 - \theta)^{29} = 1 - (1 - \theta)^{29}(1 + 29\theta). \end{aligned}$$

Function $\pi_C(\theta)$ is strictly increasing (see Figure 12.1) from $\pi_C(0) = 0$ to $\pi_C(1) = 1$, and its highest value on the set Θ_0 (null hypothesis) is attained at $\theta = 0.03$, and equals

$$\pi_C(0.03) = \max_{\theta \in \Theta_0} \pi_C(\theta) = 0.227.$$

The supermarket manager would probably be content with such a procedure: if the alternative is true, that is, if the fraction θ of unacceptable fruits exceeds 0.03, then he has at least a 22.7% chance of detecting it. In fact this chance increases rather fast with θ . For instance, for $\theta = 0.05$, we have $\pi_C(\theta) = 0.446$, and for $\theta = 0.1$, we have already $\pi_C(\theta) = 0.816$.

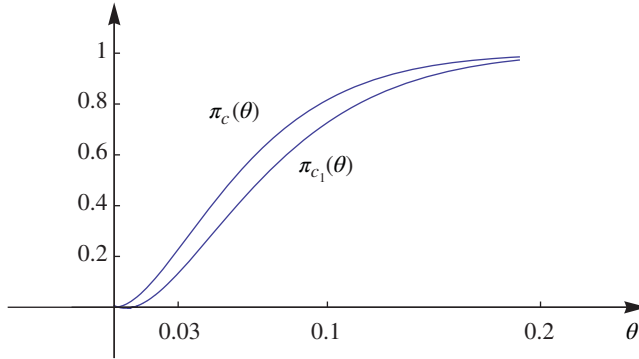


Figure 12.1 Power functions $\pi_C(\theta)$ and $\pi_{C_1}(\theta)$.

On the other hand, the supplier will not be too happy. The chances of having a shipment rejected while the standard of $\theta \leq 0.03$ is actually met can be as high as 22.7% (if θ is very close to 0.03).

■ EXAMPLE 12.6

Continuing Example 12.5, a possible solution to the problem of accepting orange shipments is to change the procedure and reject the shipment if three or more (rather than two or more) fruits are unacceptable. The critical region, C_1 , for rejecting the claim $\theta \leq 0.03$ is now $\{S_{30} \geq 3\}$, so

$$\begin{aligned} \pi_{C_1}(\theta) &= 1 - (1 - \theta)^{30} - 30\theta(1 - \theta)^{29} - 435\theta^2(1 - \theta)^{28} \\ &= 1 - (1 - \theta)^{28}(1 + 28\theta + 406\theta^2). \end{aligned}$$

Again, this is an increasing function, and (see Figure 12.1)

$$\pi_{C_1}(0.03) = \max_{\theta \in \Theta_0} \pi_{C_1}(\theta) = 0.060.$$

This time the supplier is happier, but the manager is not. He might feel that the chance of accepting a shipment with $\theta = 0.1$, equal to $1 - \pi_{C_1}(0.1) = 0.411$, is too high.

These two examples illustrate the fact that an attempt to decrease the probabilities of one type of error by changing the critical region leads to an increase of the probabilities of error of the other type. The only way to decrease both probabilities of error at the same time is to increase the sample size.

■ EXAMPLE 12.7 Randomization

In the case of orange shipments discussed in Examples 12.5 and 12.6, one could also suggest the following procedure, which does not involve any increase of the sample size and yet leads to a procedure that may be acceptable to both the manager and the supplier.

Since rejection of the shipment if $S_{30} \geq 2$ is too favorable for the manager, while rejection if $S_{30} \geq 3$ is too favorable for the supplier, one could suggest the following procedure: if S_{30} is 0, 1, or 2, the shipment is accepted by the store; if S_{30} is 4, 5, ..., 30, the shipment is rejected by the store. However, if $S_{30} = 3$, a coin is tossed, and the shipment is accepted or rejected depending on the outcome of the toss. Actually, it is not necessary to use a fair coin. One can also activate some other random mechanism such that the probability of the shipment being rejected if $S_{30} = 3$ is some fixed number γ .

The power function $\pi(\theta) = P_\theta\{H_0 \text{ is rejected}\}$ now equals

$$\pi(\theta) = \gamma P\{S_{30} = 3\} + P\{S_{30} \geq 4\},$$

where $0 < \gamma < 1$, and it lies between power functions of the two procedures considered in Examples 12.5 and 12.6. It is possible that the manager and the supplier can negotiate a value of γ that is acceptable for both of them.

The randomized procedure described in Example 12.7 is somewhat controversial. If testing statistical hypotheses is regarded as a process aimed at establishing the truth or falsehood of some statements about the experimental situation under analysis, then indeed, declaring the truth of one hypothesis on the basis of a flip of a coin may appear appalling and absurd. However, in testing statistical hypotheses according to the original intention of Neyman and Pearson (who built the foundations of this theory), the rejection and acceptance of the hypothesis are not statements about truth and falsehood. They are intended to be the guidelines for future actions. As Neyman (1950, pp. 259–260) wrote: “The terms ‘accepting’ and ‘rejecting’ a statistical hypothesis are very convenient and are well established. It is important, however, to keep their exact meaning in mind and to discard various additional implications which may be suggested by intuition. Thus, to accept a hypothesis H means only to take an action A rather than B . This does not mean that we necessarily believe that the hypothesis H is true. Also, if the application of the rule of inductive behavior ‘rejects’ H , this means only that the rule prescribes action B and does not mean that we believe that H is false.”

In light of this interpretation, the randomization of a decision is justifiable.

■ EXAMPLE 12.8

Let us consider a random sample from the $U[0, \theta]$ distribution, where $\Theta = (0, \infty)$. Suppose that we want to test the null hypothesis $H_0 : 5 \leq \theta \leq 6$. The alternative hypothesis is $H_1 : \theta < 5 \text{ or } \theta > 6$. Thus, $\Theta_0 = [5, 6]$ and $\Theta_1 = (0, 5) \cup (6, \infty)$.

We consider two tests and determine their power. First, we know (see Example 11.44) that $T_1 = X_{n:n}$ is the sufficient statistic for θ , and we might try to base the testing procedure on it. Clearly, if $T_1 \geq 6$, we should reject the null hypothesis, since it simply cannot be true in this case. Similarly, we should reject the null hypothesis if T_1 is too small, for example, $T_1 < 4.6$. Finally, we may argue that the values of T_1 slightly below 6 are also a good indication that H_0 may be false, since we always have $T_1 < 6$. The actual thresholds will depend on n . For the critical region

$$C = \{(x_1, \dots, x_n) : T_1 < 4.6 \text{ or } T_1 > 5.9\}, \quad (12.4)$$

the power of the test is

$$\pi_C(\theta) = P_\theta\{T_1 < 4.6\} + P_\theta\{T_1 > 5.9\}, \quad (12.5)$$

where

$$P_\theta\{T_1 < 4.6\} = \begin{cases} 1 & \text{if } \theta \leq 4.6 \\ \left(\frac{4.6}{\theta}\right)^n & \text{if } \theta > 4.6. \end{cases}$$

Similarly,

$$\begin{aligned} P_\theta\{T_1 > 5.9\} &= 1 - P_\theta\{T_1 \leq 5.9\} \\ &= \begin{cases} 0 & \text{if } \theta \leq 5.9 \\ 1 - \left(\frac{5.9}{\theta}\right)^n & \text{if } \theta > 5.9. \end{cases} \end{aligned}$$

Thus,

$$\pi_C(\theta) = \begin{cases} 1 & \text{for } \theta \leq 4.6 \\ \left(\frac{4.6}{\theta}\right)^n & \text{for } 4.6 < \theta \leq 5.9 \\ \left(\frac{4.6}{\theta}\right)^n + 1 - \left(\frac{5.9}{\theta}\right)^n & \text{for } \theta > 5.9. \end{cases}$$

The graph of this power function for $n = 15$ is presented in Figure 12.2.

Let us also consider a test based on statistic $\bar{X}$. We know that $E(\bar{X}) = \theta/2$. To have a comparison with the test (12.4), consider now a test with the critical region

$$C_1 = \{(x_1, \dots, x_n) : \bar{x} < 2.30 \text{ or } \bar{x} > 2.95\}. \quad (12.6)$$

To determine the exact distribution of $\bar{X}$, while possible in principle, is very cumbersome, so we will rely on the approximation provided by the central limit theorem, using the fact that $\text{Var}(X_i) = \theta^2/12$. The distribution of $\bar{X}$ is therefore approximately normal $N(\theta/2, \theta^2/12n)$, and the power of the test (12.6) is

$$\begin{aligned} \pi_{C_1}(\theta) &= P\{\bar{X} < 2.30\} + P\{\bar{X} > 2.95\} \\ &\approx 1 - P\left\{\frac{2.30 - \theta/2}{\theta/\sqrt{12n}} \leq Z \leq \frac{2.95 - \theta/2}{\theta/\sqrt{12n}}\right\} \\ &= 1 - \left[\Phi\left(\frac{2.95 - \theta/2}{\theta/\sqrt{12n}}\right) - \Phi\left(\frac{2.30 - \theta/2}{\theta/\sqrt{12n}}\right)\right]. \end{aligned}$$

The graph of this function, for $n = 15$, is presented in Figure 12.2.

As already mentioned, the expectation of a “good” test is that it should have the power as low as possible for the values of θ in the null hypothesis, and as high as possible for θ in the

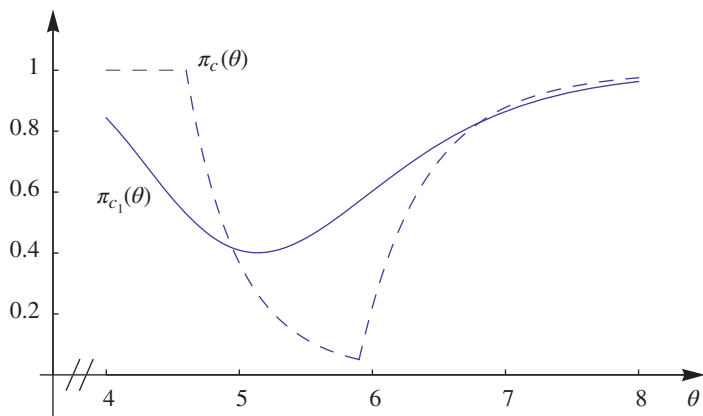


Figure 12.2 Power functions of tests C and C_1 .

alternative hypothesis. A glance at Figure 12.2 shows that the test C , based on the sufficient statistic, is much better than the test C_1 based on the sample mean $\bar{X}$.

As can be seen, for all θ in the null hypothesis H_0 , we have $\pi_C(\theta) < \pi_{C_1}(\theta)$. Actually,

$$\sup_{\theta \in \Theta_0} \pi_C(\theta) = \max(\pi_C(5), \pi_C(6)) = \max(0.2863, 0.2414) = 0.2863.$$

On the other hand,

$$\sup_{\theta \in \Theta_0} \pi_{C_1}(\theta) = \max(\pi_{C_1}(5), \pi_{C_1}(6)) = \max(0.4094, 0.6033) = 0.6033.$$

For the alternative hypothesis, when we want the power to be as high as possible, we have $\pi_C(\theta) > \pi_{C_1}(\theta)$, except for two rather narrow ranges immediately to the left of $\theta = 5$ and to the right of $\theta = 6$. The test based on critical region C may be considered “better” than the test based on critical region C_1 .

■ EXAMPLE 12.9

Finally, consider one of the classical examples, a test of hypotheses about the mean in a normal distribution with known variance. We assume that $X_1, \dots, X_n$ is a random sample from the $N(\mu, 1)$ distribution (assumption that $\sigma = 1$ is not a restriction of generality, as it simply amounts to choosing the appropriate unit for X_i 's). Suppose that we want to test the null hypothesis $H_0 : \mu = \mu_0$ against the alternative $H_1 : \mu \neq \mu_0$. In this case, the null hypothesis is simple while the alternative is composite.

The procedure will be based on the sample mean $\bar{X}$. If this average deviates “too much” from μ_0 , then the null hypothesis should be rejected. This means taking the critical region of the form

$$C = \{(x_1, \dots, x_n) : |\bar{X} - \mu_0| > k\}, \quad (12.7)$$

where k is some constant.

Recalling that $\text{Var}(\bar{X}) = 1/2$, the power function of this test is

$$\begin{aligned} \pi_C(\mu) &= P_\mu\{|\bar{X} - \mu_0| > k\} = 1 - P_\mu\{|\bar{X} - \mu_0| \leq k\} \\ &= 1 - P_\mu\{\mu_0 - k \leq \bar{X} \leq \mu_0 + k\} \end{aligned}$$

$$\begin{aligned}
 &= 1 - P \left\{ \frac{-k + \mu_0 - \mu}{1/\sqrt{n}} \leq \frac{\bar{X} - \mu}{1/\sqrt{n}} \leq \frac{k + \mu_0 - \mu}{1/\sqrt{n}} \right\} \\
 &= 1 - P \left\{ \frac{-k + \mu_0 - \mu}{1/\sqrt{n}} \leq Z \leq \frac{k + \mu_0 - \mu}{1/\sqrt{n}} \right\}.
 \end{aligned}$$

If H_0 is true, then $\mu = \mu_0$ and the power equals

$$\pi_C(\mu_0) = 1 - P\{|Z| \leq k\sqrt{n}\},$$

a quantity that depends on the chosen threshold k . Also, for any fixed $k > 0$, we have $\pi_C(\mu_0) \rightarrow 0$ as $n \rightarrow \infty$. On the other hand, for fixed $k > 0$ and n , we have $\pi_C(\mu) \rightarrow 1$ when $\mu \rightarrow \pm\infty$.

This means that for a fixed sample size n we can choose the critical region so that the probability of the type I error is equal to any preassigned level. The probability $1 - \pi_C(\mu)$ of the type II error approaches 0 as μ moves away from the null hypothesis (see Figure 12.3). How quickly it happens (i.e., how fast the power function approaches 1 as $|\mu - \mu_0|$ increases) depends on n .

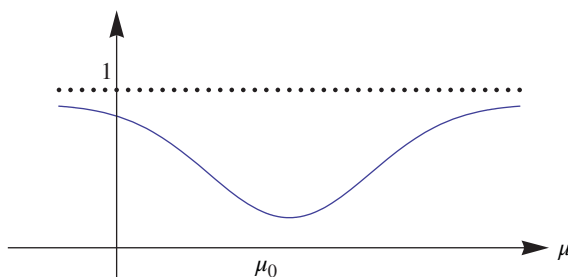


Figure 12.3 Power of the two-sided test C .

The main issues raised by the examples of this section may be summarized as follows:

1. A test, being a procedure that ultimately leads to a choice between two hypotheses, is equivalent to specifying a critical region C in the space of observations.
2. The performance of the test C is described by the power function defined as $\pi(\theta) = P\{\text{rejecting } H_0 \text{ if } \theta \text{ is the true parameter value}\}$. All criteria of choice of a test should ultimately be expressed in terms of the power function. Any such criterion should conform to the intuitive requirement that the power of a test should be as high as possible for θ in the alternative hypothesis, and as low as possible for θ in the null hypothesis.
3. The class of all possible tests (i.e., class of all possible critical regions C) is very rich, and *any* reduction of this class will facilitate the search for the best set (in whichever way the optimality is ultimately defined).

PROBLEMS

- 12.2.1** Let X_1, X_2 be a random sample from the $U[\theta, \theta + 1]$ distribution. In the test of $H_0 : \theta = 0$ against $H_1 : \theta > 0$, H_0 is rejected when $X_1 + X_2 > k$. Find the power function of the test that has probability of the type I error equal 0.05.

- 12.2.2** Consider the following procedure for testing the hypothesis $H_0 : p \geq 0.5$ against the alternative $H_1 : p < 0.5$ in $\text{BIN}(10, p)$ distribution. We take observation X_1 , and reject H_0 if $X_1 = 0$ or accept H_0 if $X_1 \geq 9$; otherwise, we take another observation X_2 (with the same distribution as X_1 and independent of it). Then, we accept or reject H_0 depending on whether $X_1 + X_2 \geq 5$ or $X_1 + X_2 < 5$. Determine and graph the power function of this procedure.
- 12.2.3** Consider three tests C_1, C_2 , and C_3 of the same hypothesis, performed independently (e.g., for each of these tests the decision is based on a different sample). Consider now the following three procedures:
 A: Reject H_0 only if all three tests reject it; otherwise, accept H_0 ,
 B: Reject H_0 only if at least two tests reject it; otherwise, accept H_0 ,
 C: Reject H_0 only if at least one test rejects it; otherwise, accept H_0 .
 (i) Express the power functions of procedures A, B, and C through power functions of tests C_1 – C_3 . (ii) Assuming that power functions of tests C_1 – C_3 are the same ($\pi_{C_1}(\theta) = \pi_{C_2}(\theta) = \pi_{C_3}(\theta) = \pi(\theta)$), graph the power functions of procedures A, B, and C.
- 12.2.4** Let X_1, X_2 be a random sample of size 2 from the $U[0, \theta]$ distribution. We want to test $H_0 : \theta = 3$ against $H_1 : \theta = 2$ (observe that H_0 and H_1 do not exhaust all possibilities). (i) H_0 will be rejected if $\bar{X} < c$. Find c such that the probability of a type I error of the test is $\alpha = 0.05$ and determine its power function. (ii) Answer (i) if H_0 will be rejected when $X_{2:2} < c$. (iii) Compare the power functions of both tests.
- 12.2.5** An urn contains five balls, r red and $5 - r$ white. The null hypothesis states that all balls are of the same color (i.e., $H_0 : r = 0$ or $r = 5$). Suppose that we take a sample of size 2 and reject H_0 if the balls are of different colors. Find the power of this test for $r = 0, \dots, 5$ if the sample is drawn: (i) Without replacement. (ii) With replacement. (iii) In each case find the probability of a type I error. (iv) Answer (i)–(iii) if the null hypothesis is now $H_0 : r = 1$ or $r = 2$, and it is rejected when all k balls selected are white.
- 12.2.6** A large box is full of chips in three colors: red, black, and green. The hypothesis to be tested is that chips in the box are mixed in equal proportions with the alternative that they are not. Four chips are to be selected with replacement and the null hypothesis will be rejected if at least one color of chips will be missing. Determine the power of the test if chips in the box are mixed in proportions 0.25 (red), 0.25 (black), and 0.5 (green).
- 12.2.7** Let $X_1, \dots, X_9$ be a random sample from the $N(\mu, 1)$ distribution. To test the hypothesis $H_0 : \mu \leq 0$ against $H_1 : \mu > 0$, one uses the test “reject H_0 if $3 \leq \bar{X} \leq 5$.” Find the power function and show that this is a bad test.
- 12.2.8** A coin is thrown independently 40 times to test that the probability of heads θ is 0.5 against the alternative that it is greater than 0.5. The test rejects the null hypothesis if more than 24 heads are observed in these 40 tosses. (i) Use the CLT with continuity correction to approximate the power function formula. (ii) Obtain the size of the test, and the power for $\theta = 0.6$.

12.3 MOST POWERFUL TESTS

If in the parametric situation considered in this chapter both hypotheses, null and alternative, are simple, it means that we are testing $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$ where θ_0 and θ_1 are two parameter values.

To simplify the notation, let us write $f(x; \theta_0) = f_0(x)$ and $f(x; \theta_1) = f_1(x)$. The only assumption about f_0 and f_1 is that they represent different probability distributions. In the discrete case, this means that $P\{X = x|H_0\} = f_0(x) \neq f_1(x) = P\{X = x|H_1\}$ for some x . In the continuous case, f_0 and f_1 are densities, and it is not enough that f_0 and f_1 differ at some isolated point, or even on a countable set of points. Thus in the continuous case, we assume that

$$\int_A f_0(x) dx \neq \int_A f_1(x) dx$$

for some set A .

We present the motivation for steps of the proof of the randomization principle theorem that will be introduced later. We consider the case of a single observation, but the extension to the case of n observations will be obvious.

Suppose that we determined some critical region C . Thus, the test is

$$\text{reject } H_0 \quad \text{if } x \in C. \quad (12.8)$$

Since both hypotheses are simple, we can determine the probabilities of errors of both types. Thus, the probability of rejecting H_0 if it is true (type I error) equals

$$\alpha = P\{x \in C|H_0\} = \begin{cases} \int_C f_0(x) dx \\ \sum_{x \in C} f_0(x), \end{cases} \quad (12.9)$$

depending on whether we deal with a continuous or a discrete case. Similarly, the probability of not rejecting H_0 if H_1 is true (type II error) is

$$\beta = P\{x \notin C|H_1\} = \begin{cases} 1 - \int_C f_1(x) dx \\ 1 - \sum_{x \in C} f_1(x). \end{cases} \quad (12.10)$$

Since to decrease β we must increase the integral in (12.10), the problem is to choose C so as to maximize $\int_C f_1(x) dx$. It will be convenient to partition the set of values of x into four sets A_0, A_1, A_2, A_3 , depending on which of the two densities is zero and which is positive (see Figure 12.4).

Let us first consider the set

$$A_1 = \{x : f_0(x) = 0, f_1(x) > 0\}. \quad (12.11)$$

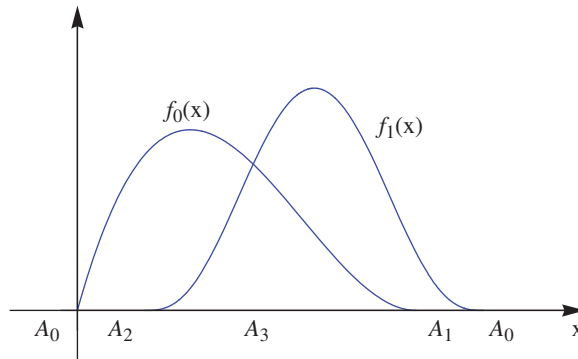


Figure 12.4 Partition into sets A_0, A_1, A_2, A_3 .

If any part of this set lies outside C , then we can improve the test by including it into C , that is, take the test with critical region $C^* = C \cup A_1 = C \cup (C^c \cap A_1)$.

Indeed, for the test C^* (in the continuous case), we have

$$\alpha^* = \int_{C^*} f_0(x) dx = \int_C f_0(x) dx + \int_{C^c \cap A_1} f_0(x) dx = \alpha,$$

since $f_0(x) \equiv 0$ on $C^c \cap A_1$. Thus, test C^* has the same probability of a type I error as test C . However (again for the continuous case), we have

$$\begin{aligned} \beta^* &= 1 - \int_{C^*} f_1(x) dx = 1 - \int_C f_1(x) dx - \int_{C^c \cap A_1} f_1(x) dx \\ &\leq 1 - \int_C f_1(x) dx = \beta, \end{aligned}$$

which means that the probability of a type II error has decreased. This argument shows that the set A_1 should be totally contained in the critical region C .

On the other hand, consider the set

$$A_2 = \{x : f_0(x) > 0, f_1(x) = 0\}. \quad (12.12)$$

A reasoning analogous to that carried for A_1 shows that the entire set A_2 should lie outside the critical region C . Next, the set

$$A_0 = \{x : f_0(x) = f_1(x) = 0\}$$

plays no role in our reasoning. The way in which this set is partitioned between C and its complement has no effect on α and β .

Now let

$$A_3 = \{x : f_0(x) > 0 \text{ and } f_1(x) > 0\}. \quad (12.13)$$

The problem reduces to finding the best way of partitioning A_3 between C and its complement, in order to improve C .

To simplify the argument, consider the discrete case. Suppose that we have already selected some set C as a candidate for a critical set, and that $C \supset A_1$ and $A_2 \subset C^c$.

Let $x^* \notin C$ (with $x^* \in A_3$), and consider the consequences of changing C by including x^* in it. The probability of a type I error will change to

$$\sum_{x \in C} f_0(x) + f_0(x^*) = \alpha + f_0(x^*) > \alpha.$$

On the other hand, the probability of a type II error will change to

$$1 - \left[\sum_{x \in C} f_1(x) + f_1(x^*) \right] = \beta - f_1(x^*) < \beta.$$

Thus, α increased by $f_0(x^*)$ while β decreased by $f_1(x^*)$.

In a similar way, suppose that $x^{**} \in C$ and that we remove it from C . Then, α will decrease by $f_0(x^{**})$, while β will increase by $f_1(x^{**})$.

Consequently, if $x^* \notin C$ and $x^{**} \in C$, switching the role of these points results in the following change:

$$\begin{aligned} \alpha &\text{ changes to } \alpha + (f_0(x^*) - f_0(x^{**})), \\ \beta &\text{ changes to } \beta + (f_1(x^{**}) - f_1(x^*)). \end{aligned}$$

Since we want to minimize both α and β , such a change should *always* be carried out if $f_0(x^*) - f_0(x^{**}) \leq 0$ and $f_1(x^{**}) - f_1(x^*) \leq 0$, with at least one inequality being strict.

The inequalities above mean (remembering that x^* and x^{**} are in A_3) that $f_1(x^{**}) \leq f_1(x^*)$ and $f_0(x^{**}) \geq f_0(x^*) > 0$. So since one inequality is strict

$$\frac{f_1(x^{**})}{f_0(x^{**})} < \frac{f_1(x^*)}{f_0(x^*)}. \quad (12.14)$$

We therefore obtained the following principle:

Reduction Principle In choosing the critical regions, one should restrict the considerations to sets based on the *likelihood ratio* $f_1(x)/f_0(x)$ of the form

$$C_k = \left\{ x : \frac{f_1(x)}{f_0(x)} \geq k \right\}. \quad (12.15)$$

□

Indeed, if a critical set is not of the form (12.15), then there exist points x^* and x^{**} , with $x^* \notin C$ and $x^{**} \in C$ and such that (12.14) holds. Then, a better critical region can be obtained by exchanging the roles of x^* and x^{**} .

It should be noticed that we did not solve the problem of finding the best test. In fact we did not even specify the criterion to be optimized. The reduction principle above tells us only which tests should not be used—in other words—it specifies the class of tests from which the choice should be made, provided only that the optimality criterion is compatible with the general motive to decrease *both* probabilities of errors.

Two obvious ways of defining the criterion to be optimized are as follows:

1. Impose an upper bound on one of the probabilities of errors, and minimize the probability of the other kind of error.
2. Minimize a linear combination of the two error probabilities.

These two approaches are closely related, the first being used in the original Neyman–Pearson lemma dating back to 1933. In formulating the theorem below, we will consider a general situation of a random sample $\mathbf{X} = (X_1, \dots, X_n)$ from one of the two distributions f_0 or f_1 . The null hypothesis asserts that the distribution is f_0 . For $i = 0, 1$ we will write $f_i(\mathbf{x}) = f_i(x_1) \times \dots \times f_i(x_n)$. The critical regions under consideration are now subsets of the n -dimensional Euclidean space.

Theorem 12.3.1 (Neyman–Pearson Lemma) *Let C^* be a critical region that has the following properties: There exists a constant $k > 0$ such that:*

- (a) *If $f_1(\mathbf{x})/f_0(\mathbf{x}) > k$, then $\mathbf{x} \in C^*$.*
- (b) *If $f_1(\mathbf{x})/f_0(\mathbf{x}) < k$, then $\mathbf{x} \notin C^*$ with points at which $f_1(\mathbf{x})/f_0(\mathbf{x}) = k$ partitioned between C^* and its complement in some way.*

Let C be any critical region. Then, $\alpha(C) \leq \alpha(C^)$ implies $\beta(C) \geq \beta(C^*)$, and if $\alpha(C) < \alpha(C^*)$, then $\beta(C) > \beta(C^*)$.*

Proof: Without danger of confusion, we can use the symbols A_0, A_1, A_2 , and A_3 for the four sets in $\mathbf{R}^n$, depending on which of the joint densities $f_0(\mathbf{x})$ and $f_1(\mathbf{x})$ is zero and which is strictly positive. If we assume that $c/0 = \infty$ for any $c > 0$, then the critical region C^* in the lemma is such that

$$A_1 \subset C^*, \quad A_2 \subset (C^*)^c. \quad (12.16)$$

Indeed, we have $f_1(\mathbf{x})/f_0(\mathbf{x}) = \infty > k$ on A_1 and $f_1(\mathbf{x})/f_0(\mathbf{x}) = 0 < k$ on A_2 . The theorem follows from the argument for the one-dimensional case preceding this proof: if C does not

meet conditions (12.16), then one or both of its error probabilities can be improved, as stated in the assertion of the Neyman–Pearson lemma.

Next, the set A_0 plays no role, and the question remains about points $\mathbf{x}$ in A_3 . In the discrete case, the argument given for one-dimensional case remains valid: if C is not of the form specified in the lemma, then there exist $\mathbf{x}^* \notin C$ and $\mathbf{x}^{**} \in C$ such that (12.15) holds, and by switching these points we improve both error probabilities. It remains therefore to prove the theorem in the continuous case. Consider the sets

$$C^* \cap C^c = \left\{ \mathbf{x} : \frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} > k, \mathbf{x} \notin C \right\}$$

and

$$(C^*)^c \cap C = \left\{ \mathbf{x} : \frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} < k, \mathbf{x} \in C \right\}.$$

We have here

$$\begin{aligned} \alpha(C^*) - \alpha(C) &= \int_{C^*} f_0(\mathbf{x}) d\mathbf{x} - \int_C f_0(\mathbf{x}) d\mathbf{x} \\ &= \int_{C^* \cap C^c} f_0(\mathbf{x}) d\mathbf{x} - \int_{C \cap (C^*)^c} f_0(\mathbf{x}) d\mathbf{x} \\ &\leq \int_{C^* \cap C^c} \frac{1}{k} f_1(\mathbf{x}) d\mathbf{x} - \int_{C \cap (C^*)^c} \frac{1}{k} f_1(\mathbf{x}) d\mathbf{x} \\ &= \frac{1}{k} \int_{C^c} f_1(\mathbf{x}) d\mathbf{x} - \frac{1}{k} \int_{(C^*)^c} f_1(\mathbf{x}) d\mathbf{x} \\ &= \frac{1}{k} [\beta(C) - \beta(C^*)]. \end{aligned}$$

The inequality between the extreme terms completes the proof: if $\alpha(C) \leq \alpha(C^*)$, then the left-hand side is nonnegative and so must be the right-hand side, which means that $\beta(C) \geq \beta(C^*)$. If the left-hand side is strictly positive, so must be the right-hand side. $\square$

We will now formulate the analogue of the Neyman–Pearson lemma in the case where the criterion to be minimized is a linear combination of the two error probabilities. The proof, which follows closely the reasoning in the proof of Neyman–Pearson lemma, will be omitted.

Theorem 12.3.2 *Suppose that in testing $H_0 : f = f_0$ against $H_1 : f = f_1$, it is desired to find the critical region C^* in n -dimensional space, such that for any other critical region C we have*

$$A\alpha(C^*) + B\beta(C^*) \leq A\alpha(C) + B\beta(C),$$

where $A > 0$, $B > 0$ are given constants. Then C^* contains all points $\mathbf{x}$ such that $f_1(\mathbf{x})/f_0(\mathbf{x}) > A/B$, and its complement contains all points $\mathbf{x}$ such that $f_1(\mathbf{x})/f_0(\mathbf{x}) < A/B$. The points where $f_1(\mathbf{x})/f_0(\mathbf{x}) = A/B$ can be allocated between C^* and its complement in an arbitrary way.

Since in the case of simple hypotheses the power of the test with critical region C is $1 - \beta(C)$, the Neyman–Pearson lemma gives us in effect a rule of constructing the *most powerful* test, with the preassigned probability $\alpha(C)$ of type I error (as we will see, this probability is determined by the choice of a constant k).

We now introduce three important definitions. Let C be the critical region for testing the null hypothesis $H_0 : \theta \in \Theta_0$ against the alternative $H_1 : \theta \in \Theta_1$, and let $\pi_C(\theta) = P_\theta\{\mathbf{X} \in C\}$ be the power of the test.

Definition 12.3.1 The *size* of the test C is defined as

$$\bar{\alpha}(C) = \sup_{\theta \in \Theta_0} \pi_c(\theta). \quad \square$$

Thus, the size of the test of a simple null hypothesis is the probability of a type I error, while for the composite null hypotheses the size is the least upper bound for all probabilities of type I errors.

Definition 12.3.2 Any number α satisfying the inequality $\bar{\alpha}(C) \leq \alpha$ is called the *level* of test C , and a test C satisfying $\bar{\alpha}(C) \leq \alpha$ will be called an α -*level* test. $\square$

According to this definition, a test with $\bar{\alpha}(C) = 0.01$ is a 1%-level test, as well as a 5%-level test, and so on.

Finally, we define a significance level. This is not so much a property of a test as a constraint imposed by the statistician.

Definition 12.3.3 If in testing the hypothesis H_0 the statistician chooses a number α_0 ($0 < \alpha_0 < 1$) and decides to use only those tests C whose size satisfies the inequality $\bar{\alpha}(C) \leq \alpha_0$, then α_0 is called the *significance level* chosen by the user. $\square$

It is often felt that a report that a given hypothesis was rejected at the significance level $\alpha = 0.05$ is not informative enough (e.g., the hypothesis might also be rejected at the significance level $\alpha = 0.01$ or even lower). Thus, it is customary to report the so called *p-value*, defined as the lowest level at which the null hypothesis would be rejected by the test. More intuitively, the *p-value* of the results actually observed is the probability that if the experiment were repeated, we would obtain results that give at least as strong evidence for the alternative hypothesis (or equivalently, as strong evidence against the null hypothesis) as the present result. For instance, suppose that we use statistic T for testing, and we reject the null hypothesis if T is large. The result which we observe is $T = t_0$, say. Then, the *p-value* of this result is $P\{T \geq t_0 | H_0\}$, so that the smaller is the *p-value*, the stronger is the evidence that suggests rejection of the null hypothesis.²⁶

In the case of two-sided tests based on statistic T with a symmetric distribution, the *p-value* of the result t_0 is defined as $2P(T > |t_0|)$. Here the rationale is that the observations more “in favor” of the alternative hypothesis are those above $|t_0|$ and below $-|t_0|$. In the case of tests based on a statistic that does not have a symmetric distribution, the ideas behind the concept of *p-value* become rather fuzzy, and there is no definition on which all statisticians agree. Consequently, one may find various “competing” definitions in different texts. For a review of these definitions, see Gibbons and Pratt (1975).

An important comment is needed here. A *p-value* is a statistical index that can be useful in understanding the results of data analysis, but it should be used carefully since it has been frequently misinterpreted or even misunderstood in quantitative research. The *p-value*’s interpretation often became just a simple comparison with a 0.05 threshold. The conclusions about the null hypothesis being either “true” or “false” have been made on the basis of the side of the 0.05 threshold on which the *p-value* is. It has to be understood that the *p-value* is not a statement about the null hypothesis, but rather it provides the evidence about the fit of the actual data to the model specified by the null hypothesis. Moreover, the *p-value* does

²⁶Very high *p-values* (close to 1) might indicate that the data were manipulated so as to make them “more conforming” to the null hypothesis. The rationale here is as follows: Suppose that the *p-value* of the data is 0.999. This means that the “fit” to H_0 is so perfect that only once in a 1,000 times we would observe a better fit. The same principles which lie at the foundation of testing allow us to reach the conclusion that the data were fudged to fit H_0 .

not measure the size or the importance of the effect being investigated since it is affected by the sample size. Large samples can yield a small p -value even though the effect is really little while one could get a large p -value when the effect is substantial only because the sample size might not be large enough.

The p -value, as any other single index, cannot adequately summarize the results of the experiment. This is why statisticians have been proposing other approaches for interpretation of research data, such as estimation and its confidence and prediction intervals, or Bayesian credibility intervals, to mention just a few. The common misuse of p -values received a lot of attention in recent years and finally started a vivid discussion that engaged many statistical experts. We refer interested readers to the American Statistical Association statement on the use of the p -value (Wasserstein and Lazar, 2016), and papers in the special February 2019 issue (Supplement 1) of *The American Statistician*, with over 40 papers on the topic.

We will now illustrate the procedure of test selection by some examples. In the four initial examples we will also illustrate another aspect: how an empirical hypothesis becomes “translated” into a statistical hypothesis.

■ EXAMPLE 12.10

Suppose that a scientist believes he found the exact location where memory is stored in the rats’ brain and needs to confirm it by an experiment. He genetically engineers 10 rats so that the brain region in question can be remotely inactivated by shining light onto the brain. The rats are then trained to find food in a maze. In running through the maze, the rat makes three binary choices and as a result ends in one of eight final locations, of which only one contains food. Then, the scientist turns on the light to turn off the part of the brain that—according to the researcher’s hypothesis—stores the acquired knowledge. The rats run the maze again, and the experimenter observes the number X of rats that reach the arm with food. Without any memory left, each rat has probability $(1/2)^3 = 0.125$ of finding the way to the food on the first trial. Suppose that—according to the experimenter—if this probability is as high as 0.3, it means that part of the memory must be stored in another (uninhibited) region of the brain. If the rats run through the maze independently, and each has the same probability p of choosing the path leading to food, then X has a binomial distribution $\text{BIN}(10, p)$. We have now two hypotheses: one asserting that $p = 0.125$ (if the researcher is correct in identifying the memory storage region of the brain), and the other asserting that $p = 0.3$ (if the researcher is wrong). Suppose that from the researcher’s point of view the error of rejecting the hypothesis $p = 0.125$ (i.e., no memory is left), if in fact it is true, is more serious than accepting $p = 0.125$ if in fact $p = 0.3$.

In this case, we set the null hypothesis as $H_0 : p = 0.125$ and the alternative is $H_1 : p = 0.3$. Consequently, $f_0(x) = \binom{10}{x} (0.125)^x (0.875)^{10-x}$ and $f_1(x) = \binom{10}{x} (0.3)^x (0.7)^{10-x}$. According to the Neyman–Pearson lemma, we should choose the critical region of the form $\{x : f_1(x)/f_0(x) \geq k\}$ for some k . But

$$\frac{f_1(x)}{f_0(x)} = \frac{(0.3)^x (0.7)^{10-x}}{(0.125)^x (0.875)^{10-x}} = 0.107 \cdot 3^x.$$

The inequality $\{x : f_1(x)/f_0(x) \geq k\}$ is equivalent to the inequality $x \geq k^*$ for some k^* . This means that the critical region C is formed of the right tail of the distribution of X . Since the possible values of X are 0, 1, ..., 10, we must simply determine the smallest value that belongs to C . Here the choice depends on the significance level α_0 of the test. Suppose that the experimenter decides to use $\alpha_0 = 0.05$. Then k^* must be such that $P\{X \geq k^* | p = 0.125\} \leq 0.05$; that is, $P\{X < k^* | p = 0.125\} \geq 0.95$. We have here $P\{X < 4 | p = 0.125\} = 0.973$ while $P\{X < 3 | p = 0.125\} = 0.881$. It

follows that we must reject the null hypothesis $H_0 : p = 0.125$ if $X \geq 4$; that is, if four or more rats find their way to the food on the first try. This test has the probability of a type I error equal to $1 - 0.973 = 0.027$ and the probability of a type II error equal $P\{X < 4|p = 0.3\} = 0.649$.

Some comments about this example appear to be in order. First, we see that the type II error has rather high probability, whereas the probability of the type I error is below the level of significance. A test with a critical region $C = \{3, 4, \dots, 10\}$ will have $\alpha(C) = 0.119$, while $\beta(C) = 0.382$. It is clear that among tests satisfying the assertion of the Neyman–Pearson lemma, none has the probability of a type I error equal to the desired significance level 0.05. This is due to the discrete nature of the test statistic X .

A procedure with the probability of a type I error $\alpha_0 = 0.05$ exists if we allow randomized procedures. That is, suppose that we decide to reject H_0 if $X \geq 4$, accept it if $X \leq 2$, and activate some auxiliary random mechanism and reject H_0 with probability γ if $X = 3$. Then the probability of rejecting H_0 if it is true is

$$\gamma P\{X = 3|p = 0.125\} + P\{X \geq 4|p = 0.125\} = 0.092\gamma + 0.027,$$

which equals 0.05 if $\gamma = 0.25$.

A procedure that attains a significance level $\alpha_0 = 0.05$ is therefore as follows: If four or more rats reach food on their first trial, then reject H_0 . If three rats reach food on the first trial, then toss two coins and reject H_0 if both show up heads. In all other cases accept H_0 .

As was already pointed out, such a randomized procedure may be suggested and agreed upon in the process of acceptance or rejection of merchandise. However, in the case of a decision regarding scientific hypotheses, it would be disturbing to rely on the toss of a coin.

A way out from the dilemma is to abandon the concept of a significance level as a quantity imposed by the experimenter. One can then proceed to minimize a linear combination of errors of type I and type II as in the Theorem 12.3.2. A procedure that attains it never requires randomization.

■ EXAMPLE 12.11

Continuing Example 12.10, if we want to find a critical region C so as to minimize the linear combination $10\alpha(C) + \beta(C)$, we must include in the critical region all x such that $f_1(x)/f_0(x) > 10$, which means that

$$\frac{(0.3)^x(0.7)^{10-x}}{(0.125)^x(0.875)^{10-x}} > 10$$

or $3^x > 93.46$; hence $x > 4.13$. Then, if the type I error is considered 10 times as serious as the type II error, the null hypothesis should be rejected only if five or more rats find their way to the food on the first attempt.

■ EXAMPLE 12.12

Suppose that we have a well with drinking water that is thought to be contaminated with bacteria. A fixed volume V of water from the well is sent to the laboratory for testing. Assume that the admissible norm is N bacteria of a certain kind in volume V , with $5N$ bacteria in volume V indicating an undesirable level of contamination. Here $N > 0$, so that some positive level of contamination is acceptable as safe.

The procedure in the laboratory is as follows: The water sent for analysis is thoroughly mixed, and then a sample of volume v is drawn from it, where $v \ll V$. A technician inspects the sample under the microscope and reports the number

X of bacteria that she sees. It is assumed that each of the bacteria present in the observed sample is recorded by the technician with probability π (assumed known), independently of the other bacteria.

This process is repeated n times, generating reports $X_1, X_2, \dots, X_n$ of the numbers of bacteria observed in different samples. We assume that the probability $1 - \pi$ of overlooking a bacteria is the same for all technicians.

It is desired to test the null hypothesis that the water in the well is safe against the alternative that it contains an undesirable level of bacteria. To solve this problem, it is first necessary to determine the distributions of the observable random variables and translate the empirical hypotheses stated above into statistical hypotheses.

Let $\lambda_0 = N/V$ be the density of bacteria in the well allowed by the safety standards so that the null hypothesis is $H_0 : \lambda = \lambda_0$, where λ is the actual density of bacteria per unit volume of water in the well. The alternative hypothesis is $H_1 : \lambda = 5N/V = 5\lambda_0 = \lambda_1$.

We assume now that when a sample of volume v is taken, each of the bacteria has the probability v/V of being included in the sample, independently of other bacteria. The assumption that $v \ll V$ makes the probability v/V small, and from Example 6.14 (about “thinning” of Poisson processes), it follows that in each sample of volume v the actual number of bacteria will have Poisson distribution with mean Mv/V , where M is the actual number of bacteria in volume V . Thus, under the null hypothesis H_0 , in each sample of volume v the number of bacteria will have Poisson distribution with mean $Nv/V = \lambda_0 v$, whereas under the alternative, the mean will be $\lambda_1 v = 5\lambda_0 v$.

We assume also that the total volume of water inspected by technicians, $v_1 + v_2 + \dots + v_n$, is still small as compared with V so that the Poisson approximation applies to the total number of bacteria in all samples. Finally, the counts $X_1, \dots, X_n$ (see Example 6.14) have a Poisson distribution with mean $\lambda_0 \pi$ under hypothesis H_0 and mean $\lambda_1 \pi$ under the alternative hypothesis H_1 .

Letting $\mathbf{X} = (X_1, \dots, X_n)$, we know that the test (whether most powerful, in the sense of the Neyman–Pearson lemma or minimizing the linear combination of error probabilities) is based on the likelihood ratio $f_1(\mathbf{x})/f_0(\mathbf{x})$, with values larger than a threshold leading to rejection of the null hypothesis. Denoting $S_n = X_1 + \dots + X_n$ for $i = 0, 1$, we have

$$f_i(\mathbf{x}) = \prod_{j=1}^n f_i(x_j) = \prod_{j=1}^n \frac{(\pi \lambda_i v)^{x_j}}{x_j!} e^{-\pi \lambda_i v} = \frac{(\pi v)^{S_n}}{\prod (x_i!)} e^{-n\pi \lambda_i v} \times \lambda_i^{S_n}.$$

Consequently, remembering that $\lambda_1 = 5\lambda_0$, we obtain

$$\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} = e^{-4n\pi \lambda_0 v} \times 5^{S_n},$$

so that the null hypothesis H_0 should be rejected if the total count S_n reported by all technicians exceeds some threshold k .

The actual value of k depends on the numerical values of the parameters. For instance, suppose that we take $n = 50$ samples to be inspected by technicians, and that probability π of recording bacteria by each of them is $\pi = 0.1$ (i.e., each technician records about 1 bacterium out of each 10 present). Furthermore, assume that the number of bacteria allowed is 10,000 per liter (10^3 cm^3), with the volume v taken for inspection being $1/20$ of a cubic centimeter. This gives $\lambda_0 = 10^4/10^3 = 10$ with $v = 0.05 \text{ cm}^3$, so

$$\pi \lambda_0 v = 0.1 \times 10 \times 0.05 = 0.05.$$

Under the null hypothesis H_0 , the total count $X_1 + \cdots + X_{50}$ in all samples has a Poisson distribution with mean $0.05 \times 50 = 2.5$. Under the alternative, this count is still Poisson, but with mean 12.5.

If we want to test the null hypothesis at significance level $\alpha = 0.05$, then k is determined by

$$P\{S_{50} \geq k | \lambda = 2.5\} \leq 0.05,$$

where S_n has a Poisson distribution with mean 2.5. We have $P\{S_n \leq 5\} = 0.9580$, which means that $P\{S_n \geq 6\} = 0.042$. We may take $k = 6$, so the probability of type II error of this test is $\beta = P(S_{50} < 6 | \lambda = 12.5) = 0.015$. Thus, the chances of raising a false alarm (in this case the type I error) are about 4%, whereas the chances of failing to notice a dangerous level of contamination are only about 1.5%.

■ EXAMPLE 12.13

Continuing Example 12.12, suppose that the laboratory is under a financial squeeze and decides to save on the cost of observations. The technicians are instructed to inspect each sample of size v and report only whether the bacteria were found. The saving here is that technicians stop searching the sample as soon as one bacterium is found (instead of continuing the search and counting). In a sample $Y_1, \dots, Y_n$, $Y_j = 1$ or 0 depending on whether bacteria are found in the j th sample. We have, for $i = 0, 1$,

$$P\{Y_j = 1 | H_i\} = 1 - P\{Y_j = 0 | H_i\} = 1 - e^{-\lambda_i \pi v}. \quad (12.17)$$

Consequently, the total number of samples where bacteria were found, $S = Y_1 + \cdots + Y_n$, has a binomial distribution, with the number of trials n and probability of success given by (12.17), depending on whether H_0 or H_1 is true. Recall that $\lambda_1 = 5\lambda_0$, so the likelihood ratio is now

$$\begin{aligned} \frac{f_1(s)}{f_0(s)} &= \frac{P\{S = s | H_1\}}{P\{S = s | H_0\}} = \frac{\binom{n}{s} (1 - e^{-\pi \lambda_1 v})^s (e^{-\pi \lambda_1 v})^{n-s}}{\binom{n}{s} (1 - e^{-\pi \lambda_0 v})^s (e^{-\pi \lambda_0 v})^{n-s}} \\ &= \left[\frac{1 - e^{-5\pi \lambda_0 v}}{1 - e^{-\pi \lambda_0 v}} e^{4\pi \lambda_0 v} \right]^s e^{-4\pi \lambda_0 v n}. \end{aligned}$$

Since the fraction in brackets exceeds 1, the likelihood increases with s . Again, the critical region is the set $S \geq k$, where k has to be determined from the fact that under the null hypothesis S has a binomial distribution, $\text{BIN}(n, 1 - e^{-\pi \lambda_0 v})$. In particular, the laboratory may now ask the question: How large should n be to ensure that the same significance level and the same power as in the more expensive procedure with technicians counting the bacteria. We have now $e^{-\pi \lambda_0 v} = 1 - e^{-0.05} = 0.0488$ and $e^{-\pi \lambda_1 v} = 1 - e^{-0.2} = 0.1813$. We look for n and k such that $P(S \geq k) \leq 0.05$, where $S \sim \text{BIN}(n, 0.0488)$, and $P(S' \leq k) \leq 0.015$, where $S' \sim \text{BIN}(n, 0.1813)$. Based on the central limit theorem (Theorem 9.6.2), k has to satisfy

$$P\left(Z \geq \frac{k - 0.0488n}{\sqrt{(0.0488)(0.9512)n}}\right) = 0.05,$$

$$P\left(Z \leq \frac{k - 0.1813n}{\sqrt{(0.1813)(0.8187)n}}\right) = 0.015,$$

and consequently

$$\begin{cases} k = 0.0488n + (1.645)(0.2154)\sqrt{n} \\ k = 0.1813n - (2.17)(0.3852)\sqrt{n}. \end{cases}$$

The solution, rounded to integers, is $n = 81, k = 7$. The simplified observation procedure (recording only whether or not the bacteria are present) necessitates the increase of the number of samples from 50 to 81.

It is clear that in the case of continuous random variables we can always find a test (critical region C) such that $\alpha(C)$ equals the desired significance level α_0 . Below we present examples concerning distributions other than normal, leaving the detailed presentation of various tests for the normal case to a separate section.

■ EXAMPLE 12.14

Suppose that the observations $X_1, \dots, X_n$ form a random sample from the $\text{EXP}(\lambda)$ distribution. We want to test hypothesis $H_0: \lambda = 5$ against a simple alternative $H_1: \lambda = 8$ at the significance level α_0 . The likelihood ratio is then

$$\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} = \frac{\prod_{j=1}^n (8e^{-8x_j})}{\prod_{j=1}^n (5e^{-5x_j})} = \left(\frac{8}{5}\right)^n e^{-3\sum x_i},$$

and the inequality $f_1(\mathbf{x})/f_0(\mathbf{x}) > k$ is equivalent to $\sum x_i < k^*$ for some k^* .

Here $k^* = -(1/2)[n \log(5/8) + \log k]$, but the exact relation between k and k^* is not needed to determine the test. What matters most is the direction of the inequality. The way we set the solution is that we *reject* H_0 if the likelihood ratio f_1/f_0 (with alternative density on the top) is large. Typically, this condition reduces to an inequality for some statistic (such as $x_1 + \dots + x_n$, or equivalently, $\bar{x}$). In the present case, the rejection region is the left tail (i.e., values of $x_1 + \dots + x_n$ less than a certain threshold).

To continue, we need a value k^* such that

$$P\{X_1 + \dots + X_n < k^* | H_0\} = \alpha_0.$$

Each X_i has exponential distribution with parameter λ_0 , hence with the mean $1/\lambda_0$. Consequently, $2\lambda_0 X_i$ has an exponential distribution with mean 2, that is, a chi-square distribution with 2 degrees of freedom (see Theorem 8.4.3). Therefore, $2\lambda_0(X_1 + \dots + X_n)$ has a chi-square distribution with $2n$ degrees of freedom.

In the case under consideration, we have $\lambda_0 = 5$; hence, the critical threshold k^* can be obtained from the tables of chi-square distribution with $2n$ degrees of freedom:

$$\begin{aligned} P\{X_1 + \dots + X_n < k^* | H_0\} &= P\{10(X_1 + \dots + X_n) < 10k^* | H_0\} \\ &= P\{\chi_{2n}^2 < 10k^*\} = \alpha_0. \end{aligned}$$

Taking, as an example, $n = 10$ and $\alpha_0 = 0.1$, we obtain $10k^* = \chi_{0.9,20}^2 = 12.44$, which gives $k^* = 1.244$.

The probability of a type II error is

$$\begin{aligned} P\{X_1 + \dots + X_n > 1.244 | H_1\} &= P\{16(X_1 + \dots + X_n) > 16 \times 1.244 | H_1\} \\ &= P\{\chi_{20}^2 > 19.904\} = 0.4637, \end{aligned}$$

as obtained from statistical software. This probability seems rather high as a probability of error. One reason is that the difference between $\lambda = 5$ and $\lambda = 8$ is equivalent to a small difference between means ($1/5$ and $1/8$, respectively). Another reason is that the small sample size ($n = 10$) affects the quality of the inference.

Thus far, we have considered testing the simple null hypothesis against a simple alternative in a parametric setup, where the distribution belongs to the same family $f(x; \theta)$, and the hypotheses are obtained by specifying $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$. It is important to realize that the Neyman–Pearson lemma applies to any two simple hypotheses. For instance, suppose that we have a single observation X , that comes from the $U[0, 1]$ distribution if H_0 is true or from the $\text{EXP}(3)$ distribution if H_1 is true. The densities are

$$f_0(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad f_1(x) = \begin{cases} 3e^{-3x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

The likelihood ratio equals

$$\frac{f_1(x)}{f_0(x)} = \begin{cases} 3e^{-3x} & \text{if } 0 \leq x \leq 1 \\ \infty & \text{if } x > 1, \end{cases}$$

and it is undefined for $x < 0$. Each critical region should contain the set $\{x : x > 1\}$, and also a set of the form $3e^{-3x} > k$, hence a set of the form $\{x : 0 \leq x \leq k^*\}$ for some $k^* < 1$. To determine k^* , we must have

$$P\{X \in C | H_0\} = \int_0^{k^*} f_0(x) dx + \int_1^\infty f_0(x) dx = \int_0^{k^*} f_0(x) dx = k^*.$$

It follows that if we choose the significance level $\alpha_0 = 0.05$ (say), then $k^* = 0.05$, and we reject H_0 if the observation is either below 0.05 or above 1. The power of this test is

$$1 - \beta = \int_C f_1(x) dx = 1 - \int_{0.05}^1 3e^{-3x} dx = 1 - e^{-3} + e^{-0.15} = 0.1891.$$

PROBLEMS

- 12.3.1** Let $X_1, \dots, X_{10}$ be a random sample from a $\text{POI}(\theta)$ distribution. **(i)** Find the best critical region for testing $H_0 : \theta = 0.1$ against $H_1 : \theta = 0.5$ at the significance level $\alpha = 0.05$. **(ii)** Determine the size of the test in (i).
- 12.3.2** A single observation X is taken from a $\text{BETA}(a, b)$ distribution. Find the most powerful test of the null hypothesis $H_0 : a = b = 1$, against the alternative H_1 : **(i)** $a = b = 5$, **(ii)** $a = 2, b = 3$ **(iii)** $a = b = 1/2$. Use significance level $\alpha = 0.05$.
- 12.3.3** Let X have a negative binomial distribution with parameters r and p . Find the most powerful test of $H_0 : r = 2, p = 1/2$ against $H_1 : r = 4, p = 1/2$ at significance level $\alpha = 0.05$. Find probability of type II error. Use randomized test if necessary.
- 12.3.4** Assume that X has a $N(2, \sigma^2)$ distribution. Find the best critical region for testing $H_0 : \sigma^2 = 2$ against: **(i)** $H_1 : \sigma^2 = 4$, **(ii)** $H_1 : \sigma^2 = 1$.
- 12.3.5** Let $X_1, \dots, X_n$ be a random sample from $\text{EXP}(\lambda)$ distribution. Null hypothesis $H_0 : \lambda = \lambda_0$ is tested against the alternative $H_1 : \lambda = \lambda_1$, where $\lambda_1 > \lambda_0$. Compare the power functions of the two tests: **(i)** the most powerful test and **(ii)** the most powerful test based on the statistic $X_{1:n}$. Assume that both tests have equal probabilities of a type I error.

- 12.3.6** The sample space of a test statistic X has five values: a, b, c, d, e . Test the $H_0 : f = f_0$ against $H_a : f = f_1$, where distributions f_0 and f_1 are given by the table

X	a	b	c	d	e
f_0	0.2	0.2	0.0	0.1	0.5
f_1	0.2	0.4	0.3	0.0	0.1

- 12.3.7** An urn contains six balls, r red and $6 - r$ blue. Two balls are chosen without replacement. Find the most powerful test of $H_0 : r = 3$ against the alternative $H_1 : r = 5$, with a size as close to $\alpha = 0.05$ as possible. Find the probability of a type II error for all $r \neq 3$.
- 12.3.8** A multiple-choice exam gives five answers to each of its n questions, only one being correct. Assume that a student who does not know the answer chooses randomly and is correct with probability 0.2. Let θ be the number of questions to which the student knows the answers, and let X be the number of correct responses given by this student. **(i)** Determine $f(x; \theta) = P(X = x | \theta)$. **(ii)** For $n = 50$, find the most powerful test of $H_0 : \theta = 30$ against $H_1 : \theta = 40$ at the significance level $\alpha = 0.05$. **(iii)** Determine the probability of a type II error if $\theta = 40$.
- 12.3.9** Let $X_1, \dots, X_n$ have a joint density $f(\mathbf{x}; \theta)$, and let U be a sufficient statistic for θ . Show that the most powerful test of $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$ can be expressed in terms of U .
- 12.3.10** Let $X_1, \dots, X_{10}$ be a random sample from a $\text{POI}(\lambda)$ distribution. Find: **(i)** The critical region for the most powerful test of hypothesis that $\theta = 0.7$ against the alternative that $\theta = 0.2$. Use significance level $\alpha = 0.05$. **(ii)** The size of the test in (i) and the power for $\theta = 0.2$. **(iii)** The p -value if the data are as follows: 1, 0, 0, 2, 0, 0, 0, 1, 0, 0. Interpret.

12.4 UNIFORMLY MOST POWERFUL TESTS

From the derivation of the most powerful tests in Examples 12.12 and 12.14, notice that the final form of the test is, to a large extent, independent of the choice of a specific alternative hypothesis. For instance, in Example 12.12, observations $X_1, \dots, X_n$ come from one of the two Poisson distributions. Omitting the details on size v of the water samples and the probability π of recording the presence of bacteria, we could test a simple hypothesis $H_0 : E(X_i) = \theta_0$ against a simple alternative $H_1 : E(X_i) = \theta_1$, where $\theta_1 > \theta_0$. The likelihood ratio, after canceling the factorials, turns out to be

$$\frac{f_1(\mathbf{X})}{f_0(\mathbf{X})} = \left(\frac{\theta_1}{\theta_0}\right)^{\sum X_i} \times \left(\frac{e^{-\theta_1}}{e^{-\theta_0}}\right)^n.$$

If $\theta_1 > \theta_0$, then the likelihood ratio is an increasing function of $\sum X_i$; hence, the rejection region must be of the form

$$C = \left\{ (x_1, \dots, x_n) : \sum x_i \geq k \right\}. \quad (12.18)$$

Determination of value k involves only the value of θ_0 and a level of significance α_0 ; the value θ_1 plays no role, provided that $\theta_1 > \theta_0$. We know that under the null hypothesis

$S_n = \sum_{i=1}^n X_i$ has a Poisson distribution with mean $n\theta_0$; hence,

$$k = \min \left\{ r : \sum_{j=0}^{r-1} \frac{(n\theta_0)^j}{j!} e^{-n\theta_0} \geq 1 - \alpha_0 \right\}, \quad (12.19)$$

where α_0 is the desired significance level.

This means that a test with the critical region given by (12.18) and (12.19) is the most powerful for *any* alternative hypothesis if only the reasoning leading to (12.18) applies. But the only fact about θ_1 used in the derivation is that $\theta_1 > \theta_0$. This inequality causes the ratio θ_1/θ_0 to exceed 1, and hence the likelihood ratio to increase with the sum $\sum x_i$ (for $\theta_1 < \theta_0$ the likelihood ratio would decrease with an increase of $\sum x_i$ and the critical region would comprise the left tail, i.e., we would reject H_0 if $\sum x_i \leq k$).

In the continuous case, we can restrict the considerations to tests based on critical regions. In the discrete case, one can also consider randomized procedures, that is, procedures in which the rejection or acceptance of H_0 depends both on the observation of $\mathbf{X}$ and also possibly on additional randomization.

Consequently, in the definitions below, we consider procedures for testing H_0 ; without much danger of confusion, we will use letter C for a procedure, and define its power function as

$$\pi_C(\theta) = P_\theta\{\text{procedure } C \text{ rejects } H_0\}.$$

We will fix the significance level α_0 ($0 < \alpha_0 < 1$), and let $K(H_0, \alpha_0)$ be the class of all procedures for testing H_0 whose size is at most α_0 , that is, procedures C such that

$$\sup_{\theta \in \Theta_0} \pi_C(\theta) \leq \alpha_0. \quad (12.20)$$

Definition 12.4.1 A procedure $C^* \in K(H_0, \alpha_0)$, such that

$$\pi_{C^*}(\theta) \geq \pi_C(\theta)$$

for every $\theta \in \Theta_1$ and for any $C \in K(H_0, \alpha_0)$, will be called a UMP procedure for testing H_0 against H_1 at the significance level α_0 .

In the case where C^* is nonrandomized (i.e., C^* is the critical region of a test), we will speak of a UMP test of H_0 against H_1 at the significance level α_0 . $\square$

The essence of this definition lies in the fact that *the same* test (procedure) is most powerful against *all* simple hypotheses in H_1 . The Neyman–Pearson lemma (or its extension to composite null hypothesis) asserts that the most powerful test exists against any simple alternative, but it sometimes happens that these most powerful tests are different for different simple alternatives.

■ EXAMPLE 12.15

Consider the situation of testing the simple null hypothesis $H_0 : \mu = 0$ against the composite alternative $H_1 : \mu > 0$ in case of a normal distribution $N(\mu, \sigma^2)$ with known σ^2 . Let us fix $\mu_1 > 0$ and consider the likelihood ratio test of H_0 against $H_1' : \mu = \mu_1$. We have

$$\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} = \frac{\exp\left\{-(1/2\sigma^2) \sum (x_i - \mu_1)^2\right\}}{\exp\left\{-(1/2\sigma^2) \sum x_i^2\right\}} = De^{(\mu_1/\sigma^2) \sum x_i}.$$

Since $\mu_1 > 0$, this likelihood ratio is an increasing function of $\sum x_i$ (hence also of $\bar{x}$), and the critical region of the most powerful test against $\mu > 0$ is of the form $\{\mathbf{x} = (x_1, \dots, x_n) : \bar{x} \geq k\}$.

For any given significance level α_0 , we determine k from the condition

$$P\{\bar{X} \geq k | H_0\} = P\left\{Z \geq \frac{k}{\sigma/\sqrt{n}}\right\} = \alpha_0$$

so that $k = z_{\alpha_0}\sigma/\sqrt{n}$, where z_{α_0} is the upper α_0 -quantile of the standard normal distribution.

The only assumption about μ_1 that is used here is $\mu_1 > 0$; consequently, the test with the critical region C ,

$$\text{reject } H_0 \quad \text{if } \bar{X} \geq z_{1-\alpha_0}\sigma/\sqrt{n} \quad (12.21)$$

is UMP for the alternative $H_1 : \mu > 0$.

■ EXAMPLE 12.16

Continuing Example 12.15, observe that the test (12.21) is also UMP for the composite null hypothesis $H_0 : \mu \leq 0$ against the composite alternative $H_1 : \mu > 0$. Indeed, the power of this test is

$$\begin{aligned} \pi_C(\mu) &= P_\mu\left\{\bar{X} \geq z_{\alpha_0}\frac{\sigma}{\sqrt{n}}\right\} = P\left\{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{z_{\alpha_0}(\sigma/\sqrt{n}) - \mu}{(\sigma/\sqrt{n})}\right\} \\ &= P\left\{Z \geq z_{\alpha_0} - \frac{\mu\sqrt{n}}{\sigma}\right\}. \end{aligned}$$

Consequently, for $\mu < 0$, $\pi_C(\mu) \leq P\{Z \geq z_{\alpha_0}\} = \alpha_0 = \pi_C(0)$; hence the size of the test C is α_0 .

■ EXAMPLE 12.17

Example 12.16 shows that there exists no UMP test for the hypothesis $H_0 : \mu = 0$ against the alternative $H_1 : \mu \neq 0$. Indeed, (12.21) is the most powerful test against any alternative $\mu > 0$, but it performs very poorly against the alternative $\mu < 0$; its power on such alternatives is less than α_0 . On the other hand, by symmetry, the test with critical region C' ,

$$\text{reject } H_0 \quad \text{if } \bar{X} \leq -z_{\alpha_0}\frac{\sigma}{\sqrt{n}}, \quad (12.22)$$

is most powerful against any alternative $\mu < 0$, but performs poorly against alternatives $\mu > 0$. The “compromise” test, with the critical region C'' ,

$$\text{reject } H_0 \quad \text{if } |\bar{X}| \geq z_{\alpha_0/2}\frac{\sigma}{\sqrt{n}}, \quad (12.23)$$

performs quite well for all alternatives $\mu \neq 0$ but its power is below the power of test (12.21) for $\mu > 0$, and below the power of test (12.22) for $\mu < 0$. So it is not a UMP test.

We know from the examples above that UMP tests may not exist. On the other hand, if they do exist, they provide the best available procedures (if one takes into account only the error probabilities, and not extraneous aspects, such as computation costs, etc.). It is therefore natural to ask for conditions under which UMP tests exist. The answer is given by the next theorem, which we will precede by some necessary definitions.

Definition 12.4.2 We say that the family $\{f(x; \theta), \theta \in \Theta\}$ of distributions has a *monotone likelihood ratio* in statistic T if for any two values $\theta', \theta'' \in \Theta$ with $\theta' < \theta''$, and $\mathbf{x} = (x_1, \dots, x_n)$, the likelihood ratio

$$f_n(\mathbf{x}; \theta'')/f_n(\mathbf{x}; \theta')$$

depends on $\mathbf{x}$ only through the values $T(\mathbf{x})$, and this ratio is an increasing function of $T(\mathbf{x})$. $\square$

We will now illustrate the introduced concept by some examples.

■ EXAMPLE 12.18

Consider the normal density depending on parameter μ , with σ^2 known, so that

$$f_n(\mathbf{x}, \mu) = \frac{1}{\sigma^n (2\pi)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right\}.$$

Let $\mu' < \mu''$, and for typographical reason, let $\mu' = \mu_1, \mu'' = \mu_2$. Then,

$$\begin{aligned} \frac{f_n(\mathbf{x}, \mu_2)}{f_n(\mathbf{x}, \mu_1)} &= \exp \left\{ -\frac{1}{2\sigma^2} \left[\sum (x_i - \mu_2)^2 - \sum (x_i - \mu_1)^2 \right] \right\} \\ &= \exp \left\{ -\frac{1}{2\sigma^2} \left[-2(\mu_2 - \mu_1) \sum x_i + n(\mu_2^2 - \mu_1^2) \right] \right\} \\ &= D e^{(\mu_2 - \mu_1)/\sigma^2} \sum x_i, \end{aligned}$$

where $D > 0$. Consequently, the normal distribution for a fixed σ^2 has a monotone likelihood ratio in $T = \sum X_i$, or equivalently, in $\bar{X}$.

■ EXAMPLE 12.19

Consider again the normal distribution, this time for known μ and unknown σ^2 . Let $\sigma_1^2 < \sigma_2^2$. Then,

$$\begin{aligned} \frac{f_n(\mathbf{x}, \sigma_2^2)}{f_n(\mathbf{x}, \sigma_1^2)} &= D \exp \left\{ -\frac{1}{2\sigma_2^2} \sum (x_i - \mu)^2 + \frac{1}{2\sigma_1^2} \sum (x_i - \mu)^2 \right\} \\ &= D \exp \left\{ \frac{1}{2} \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2} \right) \sum (x_i - \mu)^2 \right\}. \end{aligned}$$

Since $D > 0$ and $1/\sigma_1^2 - 1/\sigma_2^2 > 0$, we see that now the likelihood ratio is increasing in statistic $T = \sum (X_i - \mu)^2$.

■ EXAMPLE 12.20 Bernoulli Trials

Let $X_1, \dots, X_n$ be a random sample from Bernoulli distribution $f(x; p) = p^x(1-p)^{1-x}$, $x = 0, 1$, and $0 < p < 1$. For $0 < p' < p'' < 1$, we have

$$\frac{f_n(\mathbf{x}, p'')}{f_n(\mathbf{x}, p')} = \left[\frac{p''(1-p')}{p'(1-p'')} \right]^{\sum x_i} \left(\frac{1-p''}{1-p'} \right)^n.$$

Since $p''(1-p')/p'(1-p'') > 1$, the Bernoulli distribution has monotone likelihood ratio in statistic $\sum X_i$ (total number of successes in n trials).

It turns out that most of the known families of distributions have monotone likelihood ratio in some statistics. The role of families with monotone likelihood ratio for UMP tests is explained by the following theorem:

Theorem 12.4.1 *Let $\{f(x; \theta), \theta \in \Theta\}$ be a family of distributions with a monotone likelihood ratio in statistics T . Then for every α_0 , $0 < \alpha_0 < 1$, there exists a test (possibly randomized) that is UMP for testing $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$ at significance level α_0 . This test satisfies the following two conditions:*

- (a) *There exists k such that if $T(\mathbf{x}) > k$, then H_0 is rejected and if $T(\mathbf{x}) < k$, then H_0 is accepted.*
- (b) *$P_{\theta_0}\{H_0 \text{ is rejected}\} = \alpha_0$.*

Similarly, for testing $H_0 : \theta \geq \theta_0$ against $H_1 : \theta < \theta_0$, there exists k_1 such that H_0 should be rejected if $T(\mathbf{X}) < k_1$ and accepted if $T(\mathbf{X}) > k_1$.

Proof: Observe first that condition (b) may require randomization. Indeed, in view of (a) and (b), letting $\gamma = P\{H_0 \text{ is rejected} | T(\mathbf{X}) = k\}$, we must have

$$\alpha_0 = P_{\theta_0}\{T(\mathbf{X}) > k\} + \gamma P_{\theta_0}\{T(\mathbf{X}) = k\}. \quad (12.24)$$

It follows that if $P_{\theta_0}\{T(\mathbf{X}) = k\} = 0$, then we have

$$\alpha_0 = P_{\theta_0}\{T(\mathbf{X}) > k\} = P_{\theta_0}\{T(\mathbf{X}) \geq k\},$$

and the test is nonrandomized, with critical region $C = \{\mathbf{x} : T(\mathbf{x}) \geq k\}$. If $P_{\theta_0}\{T(\mathbf{X}) > k\} < \alpha_0 \leq P_{\theta_0}\{T(\mathbf{X}) \geq k\}$, then by (12.24),

$$\gamma = \frac{\alpha_0 - P_{\theta_0}\{T(\mathbf{X}) > k\}}{P_{\theta_0}\{T(\mathbf{X}) = k\}}.$$

Clearly, we have $0 < \gamma < 1$, and an auxiliary randomization with probability of success equal to γ gives a procedure with size α_0 .

Let C^* denote the procedure described by the theorem, and its power be $\pi_{C^*}(\theta) = P_{\theta}\{C^* \text{ leadstorejecting } H_0\}$. By condition (b), $\pi_{C^*}(\theta_0) = \alpha_0$.

Without loss of generality, let $\theta_1 > \theta_0$. By the Neyman–Pearson lemma, the most powerful procedure for testing the simple hypothesis $H_0^* : \theta = \theta_0$ against a simple alternative $H_1^* : \theta = \theta_1$ is based on the likelihood ratio, and rejects H_0 if $f_n(\mathbf{x}; \theta_1)/f_n(\mathbf{x}; \theta_0) > k^*$ for some k^* . But by the monotonicity of the likelihood ratio, the last condition means that $T(\mathbf{X}) > k$ for some k . Since the last condition does not depend on the choice of θ_1 , the procedure C^* described in the theorem is UMP for testing the simple hypothesis $H_0^* : \theta = \theta_0$ against the composite alternative $H_1 : \theta > \theta_0$.

We will also show that the power function of procedure C^* is nondecreasing. For any $\theta' < \theta''$ (regardless of their location with respect to θ_0), we have $\pi_{C^*}(\theta') \leq \pi_{C^*}(\theta'')$. This fact will be crucial for completing the proof.

Indeed, consider the class $\mathcal{K}$ of all procedures of testing the simple hypothesis $H_0' : \theta = \theta'$ against the simple alternative $H_1' : \theta = \theta''$ at a level of significance α , say. We know from the Neyman–Person lemma that C^* is the most powerful procedure in class $\mathcal{K}$, that is, for any $C \in \mathcal{K}$,

$$\pi_{C^*}(\theta'') \geq \pi_C(\theta''). \quad (12.25)$$

On the other hand, the (randomized) procedure C_0 : “reject $H_0^* : \theta = \theta'$ with probability α regardless of observation” satisfies the condition $\pi_{C_0}(\theta) = \alpha$ for all θ . Clearly, $C_0 \in \mathcal{K}$. So,

using (12.25), we write

$$\pi_{C^*}(\theta'') \geq \pi_{C_0}(\theta'') = \alpha = \pi_{C^*}(\theta'),$$

which shows monotonicity of $\pi_{C^*}(\theta)$.

To complete the proof, we have to show that the procedure C^* is UMP not only in the class $\mathcal{B}$ of all procedures C that satisfy the condition $\pi_C(\theta_0) \leq \alpha_0$ but also in the class $\mathcal{B}'$ of all procedures C such that $\sup_{\theta \leq \theta_0} \pi_C(\theta) \leq \alpha_0$.

Clearly, $\mathcal{B}' \subset \mathcal{B}$, and the monotonicity of the power function $\pi_{C^*}(\theta)$ shows that C^* is an element of the smaller class $\mathcal{B}'$: Indeed, for $\theta \leq \theta_0$,

$$\pi_{C^*}(\theta) \leq \pi_{C^*}(\theta_0) \leq \alpha_0.$$

This completes the proof, since we already know that for every $\theta_1 > \theta_0$, the procedure C^* maximizes the power in the larger class $\mathcal{B}$, and hence also maximizes the power in the smaller class $\mathcal{B}'$. $\square$

PROBLEMS

- 12.4.1** Check whether the following families of distributions have a monotone likelihood ratio: **(i)** Poisson. **(ii)** Exponential. **(iii)** Gamma, for each parameter separately. **(iv)** Beta, for each parameter separately.
- 12.4.2** Let $X_1, \dots, X_{16}$ be a random sample from Laplace distribution with density $f(x; \lambda) = (\lambda/2) \exp\{-\lambda|x|\}$, $\lambda > 0$. Find a UMP test for testing $H_0: \lambda = 1$ against $H_1: \lambda < 1$ at the significance level 0.01.
- 12.4.3** Let $X_1, \dots, X_n$ be a random sample from a folded normal distribution with density $f(x; \theta) = \sqrt{2/\pi}\theta \exp\{-\theta^2 x^2/2\}$, for $x > 0, \theta > 0$. **(i)** Derive the UMP test for $H_0: \theta = \theta_0$ against $H_1: \theta > \theta_0$. **(ii)** Show that the power function is increasing.
- 12.4.4** Suppose that the number of defects in magnetic tape of length t (yards) has $\text{POI}(\lambda t)$ distribution. Assume that 2 defects were found in a piece of tape of length 500 yards. **(i)** Test the hypothesis $H_0: \lambda \geq 0.02$ against the alternative $H_1: \lambda < 0.02$. Use a UMP test and the significance level $\alpha = 0.01$. **(ii)** Find the p -value and interpret. **(iii)** Find the power of the test at $\lambda = 0.015$.
- 12.4.5** The effectiveness of a standard drug in treating specific illness is 60%. A new drug was tested and found to be effective in 48 out of 70 cases when it was used. Specify an appropriate alternative hypothesis and perform the test at the 0.01 level of significance. Find the p -value.
- 12.4.6** Suppose that $X_1, \dots, X_n$ is a random sample from the $U[0, \theta]$ distribution. **(i)** Hypothesis $H_0: \theta \leq \theta_0$ is to be tested against the alternative $H_1: \theta > \theta_0$. Argue that the UMP test rejects H_0 if $X_{n:n} > c$. Find c for $\theta_0 = 5, n = 10$, and $\alpha = 0.05$. **(ii)** If the hypothesis $H_0: \theta \geq \theta_0$ is tested against $H_1: \theta < \theta_0$, show that the UMP test rejects H_0 if $X_{n:n} < c$. Find c if $\theta_0 = 5, n = 10$, and $\alpha = 0.05$.
- 12.4.7** Let $X_1, \dots, X_n$ be a random sample from the $\text{GAM}(\alpha, \lambda)$ distribution. **(i)** Derive a UMP test for the hypothesis $H_0: \alpha \leq \alpha_0$ against the alternative $H_1: \alpha > \alpha_0$ if λ is known. **(ii)** Derive a UMP test for the hypothesis $H_0: \lambda \leq \lambda_0$ against the alternative $H_1: \lambda > \lambda_0$ if α is known.
- 12.4.8** Recall Problem 12.3.8. Assume that $n = 50$ and that a student with a score at most 30 will fail. Does there exist a UMP test for the hypothesis $H_0: \theta \leq 30$? If yes, find the test; if no, justify your answer.

12.4.9 Let $X_1, \dots, X_n$ be a random sample from the distribution with $f(x; \theta) = \theta^{x-1}/(\theta+1)^x, \theta > 0, x = 1, 2, \dots$. Determine a UMP test of the hypothesis $H_0: \theta = \theta_0$ against the alternative $H_1: \theta > \theta_0$, at significance level α .

12.4.10 A reaction time to a certain stimulus (e.g., time until solving some problem) is modeled as a time of completion of r processes, running one after another in a specified order. The times $\tau_1, \dots, \tau_r$ of completion of these processes are assumed to be iid exponential with mean $1/\lambda$. If $r = 3$ and the observed reaction times of completion of these three processes (in seconds) are 15.3, 6.1, 8.5, and 9.0, test the hypothesis $H_0: \lambda \geq 0.8$ against the alternative $H_1: \lambda < 0.8$. Use $\alpha = 0.05$.

12.5 UNBIASED TESTS

As we already know, there are situations where the UMP tests do not exist. One example may be a test of hypothesis $H_0: \theta = \theta_0$ against a two-sided alternative $H_1: \theta \neq \theta_0$ (i.e., where θ_0 is not on the boundary of the parameter space Θ). Since UMP tests are highly desirable, the theoretical efforts became directed toward a reduction of the class of tests. Such reduced class might already contain a UMP test.

We present an approach that requires unbiasedness. Consider the problem of testing the null hypothesis $H_0: \theta \in \Theta_0$ against the alternative $H_1: \theta \in \Theta_1$. Let C be a testing procedure with a power function $\pi_C(\theta)$. Here the parameter space Θ may be multidimensional, and one or both of the hypotheses H_0 or H_1 may be composite.

Definition 12.5.1 The test C of H_0 against H_1 is called *unbiased* if

$$\sup_{\theta \in \Theta_0} \pi_C(\theta) \leq \inf_{\theta \in \Theta_1} \pi_C(\theta). \quad (12.26)$$

□

Since the left-hand side of (12.26) is the size of the test C , we say that C is unbiased if its power on the alternative hypothesis is never below its size. In particular, if the null hypothesis is simple, then the power function of an unbiased test reaches its minimum at θ_0 .

It turns out that in some cases where there is no UMP test in the class of all tests, there is a UMPU test.

■ EXAMPLE 12.21

Consider the case of testing the simple hypothesis $H_0: \mu = \mu_0$ against the alternative $H_1: \mu \neq \mu_0$, where observations $X_1, \dots, X_n$ are a random sample from the $N(\mu, \sigma^2)$ distribution with known σ^2 .

Intuition suggests that we take $T(\mathbf{X}) = \bar{X} - \mu_0$ as the test statistic, and reject H_0 if either $T(\mathbf{X}) < -k'$ or $T(\mathbf{X}) > k''$ for some suitably chosen positive numbers k' and k'' . To have size equal α_0 , we must choose k' and k'' so that

$$\begin{aligned} 1 - \alpha_0 &= P_{\mu_0}\{-k' < T(\mathbf{X}) < k''\} = P_{\mu_0}\{-k' < \bar{X} - \mu_0 < k''\} \\ &= P\left\{-k' \frac{\sqrt{n}}{\sigma} < Z < k'' \frac{\sqrt{n}}{\sigma}\right\}, \end{aligned}$$

that is,

$$\Phi\left(k'' \frac{\sqrt{n}}{\sigma}\right) - \Phi\left(-k' \frac{\sqrt{n}}{\sigma}\right) = 1 - \alpha_0, \quad (12.27)$$

where Φ is the CDF of standard normal random variable.

Let us investigate the power function of the test above. For the critical region $C = \{\bar{X} < \mu_0 - k'\} \cup \{\bar{X} > \mu_0 + k''\}$, we have

$$\begin{aligned}\pi_C(\mu) &= 1 - P_\mu\{-k' < \bar{X} - \mu_0 < +k''\} \\ &= 1 - P\left\{\frac{-k' + \mu_0 - \mu}{\sigma/\sqrt{n}} < Z < \frac{k'' + \mu_0 - \mu}{\sigma/\sqrt{n}}\right\} \\ &= 1 - \Phi\left(\frac{k'' + \mu_0 - \mu}{\sigma/\sqrt{n}}\right) + \Phi\left(\frac{-k' + \mu_0 - \mu}{\sigma/\sqrt{n}}\right).\end{aligned}\quad (12.28)$$

Clearly, $\pi_C(\mu)$ is a continuous differentiable function of μ , and C is unbiased if $\left.\frac{d}{d\mu}\pi_C(\mu)\right|_{\mu=\mu_0} = 0$. We have, letting φ be the density of standard normal random variable,

$$\frac{d\pi_C(\mu)}{d\mu} = \varphi\left(\frac{k'' + \mu_0 - \mu}{\sigma/\sqrt{n}}\right) \frac{\sqrt{n}}{\sigma} - \varphi\left(\frac{-k' + \mu_0 - \mu}{\sigma/\sqrt{n}}\right) \frac{\sqrt{n}}{\sigma}.$$

For $\mu = \mu_0$, we obtain

$$\left.\frac{d\pi_C(\mu)}{d\mu}\right|_{\mu=\mu_0} = \left[\varphi\left(\frac{k''}{\sigma}\sqrt{n}\right) - \varphi\left(-\frac{k'}{\sigma}\sqrt{n}\right)\right] \frac{\sqrt{n}}{\sigma},$$

which equals 0 (remembering that k' and k'' are positive) only if $k' = k''$. Thus, we obtain an unbiased test only if the two parts of the rejection regions are located symmetrically with respect to the null hypothesis value μ_0 . Formula (12.27) gives now $k' = k'' = z_{\alpha_0/2}$.

One can show that the test obtained in Example 12.21 is actually the UMPU test for hypothesis $H_0: \mu = \mu_0$ against the two-sided alternative $H_1: \mu \neq \mu_0$. It is not, however, a UMP test against one-sided alternatives (see Figure 12.5).

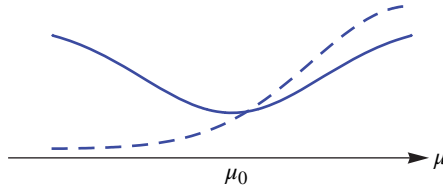


Figure 12.5 Power functions of a one-sided UMP test (dashed line) and a UMPU test (solid line).

One can argue that such a specific hypothesis like $H_0: \mu = \mu_0$ simply cannot be true. The chances that the mean is *exactly* equal to μ_0 are zero. This is true if the parameter μ is a random variable, varying from situation to situation according to some continuous prior distribution. It is also true in most cases of Bayesian priors, where μ does not vary, but the statistician's experience can be expressed in terms of a (subjective) probability distribution on μ (to be discussed in more detail in Chapter 16).

Thus, the argument goes, H_0 should be rejected at once, without any testing. As explained at the beginning of this chapter, this conclusion misses the essential intention of the theory of hypotheses testing, which is to serve not as a means of establishing the truth of hypotheses, but of establishing good rules of inductive behavior. In this sense, “accepting $H_0: \theta = \theta_0$ ” means simply that it is reasonable to act *as if* the parameter value were θ_0 , even if in fact θ is only close to θ_0 . This suggests testing a composite null hypothesis stating that θ lies in some

interval, against the alternative that it lies outside it. We will illustrate this approach with an example:

■ **EXAMPLE 12.22**

Let again $X_1, \dots, X_n$ be a random sample from the $N(\mu, \sigma)$ distribution with σ^2 known. We want to test the null hypothesis $H_0 : \mu_1 \leq \mu \leq \mu_2$ against the alternative $H_1 : \mu < \mu_1$ or $\mu > \mu_2$. To simplify the notation, let

$$\mu^* = \frac{\mu_1 + \mu_2}{2} \quad (12.29)$$

denote the midpoint between the boundaries of the null hypothesis. It appears reasonable to use the test statistic $T(\mathbf{X}) = \bar{X} - \mu^*$, and reject H_0 if $T(\mathbf{X}) < -k'$ or $T(\mathbf{X}) > k''$, where k', k'' are some positive constants. The power of this test (call it C^*) is

$$\begin{aligned} \pi_{C^*}(\mu) &= 1 - P_{\mu}\{-k' < \bar{X} - \mu^* < k''\} \\ &= 1 - P\left\{\frac{-k' + \mu^* - \mu}{\sigma/\sqrt{n}} < Z < \frac{k'' + \mu^* - \mu}{\sigma/\sqrt{n}}\right\} \\ &= 1 - \Phi\left(\frac{k'' + \mu^* - \mu}{\sigma/\sqrt{n}}\right) + \Phi\left(\frac{-k' + \mu^* - \mu}{\sigma/\sqrt{n}}\right). \end{aligned}$$

Again, to have the test unbiased, the power curve must have a minimum at μ^* , which necessitates taking $k' = k''$. Then, the size of the test (see Figure 12.6) equals to the common value of the power at the points μ_1 and μ_2 , that is, after letting $k = k' = k''$ and $\Delta = (\mu_2 - \mu_1)/2$,

$$\begin{aligned} \sup_{\mu_1 \leq \mu \leq \mu_2} \pi_{C^*}(\mu) &= 1 - \Phi\left(\frac{\mu^* - \mu_1 + k}{\sigma/\sqrt{n}}\right) + \Phi\left(\frac{\mu^* - \mu_1 - k}{\sigma/\sqrt{n}}\right) \\ &= 1 - \Phi\left(\frac{\Delta + k}{\sigma/\sqrt{n}}\right) + \Phi\left(\frac{\Delta - k}{\sigma/\sqrt{n}}\right). \end{aligned}$$

If we now require (for given μ_1, μ_2 , and σ , hence given μ^* and Δ) a test with a specified size α_0 and a specified probability β_0 of a type II error at some target value μ_t in the alternative, we can determine the threshold k and sample size n from the equations

$$\Phi\left(\frac{\Delta + k}{\sigma/\sqrt{n}}\right) - \Phi\left(\frac{\Delta - k}{\sigma/\sqrt{n}}\right) = 1 - \alpha_0$$

and

$$\Phi\left(\frac{\mu^* - \mu_t + k}{\sigma/\sqrt{n}}\right) - \Phi\left(\frac{\mu^* - \mu_t - k}{\sigma/\sqrt{n}}\right) = \beta_0.$$

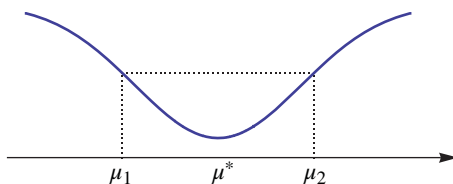


Figure 12.6 Power function of an unbiased test.

The solution has to be obtained by a numerical procedure. It can be shown that the resulting test is UMPU.

PROBLEMS

- 12.5.1** Suppose that $X_1, \dots, X_n$ is a random sample from the $U[0, \theta]$ distribution. Test hypothesis $H_0: \theta = \theta_0$ against the two-sided alternative $H_1: \theta \neq \theta_0$ using an unbiased test that rejects H_0 if $X_{n:n} < c_1$ or $X_{n:n} > c_2$, $c_1 < c_2$. Find c_1 and c_2 if $\theta_0 = 5$, $n = 10$, and $\alpha = 0.05$.
- 12.5.2** Let X be a single observation from a distribution with density $f(x, \theta) = 1 - \theta^2(x - 0.5)^3$ for $0 < x < 1$ and zero otherwise, $-1 < \theta < 1$. Find a UMPU test of $H_0: \theta = 0$ against $H_1: \theta \neq 0$.
- 12.5.3** Let $X_1, \dots, X_n$ be a random sample from the $\text{POI}(\lambda)$ distribution. Find the (approximate) UMPU test for the hypothesis $H_0: \lambda = \lambda_0$ against the two-sided alternative $H_1: \lambda \neq \lambda_0$, where λ_0 is assumed to be large. [Hint: Use the fact that if X has Poisson distribution with mean λ , then $(X - \lambda)/\sqrt{\lambda}$ converges (as $\lambda \rightarrow \infty$) in the distribution to a standard normal random variable.]
- 12.5.4** Assume that $X_1, \dots, X_n$ is a random sample from the $N(0, \sigma^2)$ distribution. For testing hypothesis $H_0: \sigma = \sigma_0^2$ against the alternative $H_1: \sigma^2 \neq \sigma_0^2$ at significance level α , find an unbiased test with a critical region of the form “reject H_0 if $\sum_{i=1}^n X_i^2/\sigma_0^2 < C_1$ or $\sum_{i=1}^n X_i^2/\sigma_0^2 > C_2$.”
- 12.5.5** Let $X_1, \dots, X_n$ be a random sample from the $N(\mu, \sigma^2)$ distribution with both parameters unknown. Find an unbiased test with a critical region of the form “reject H_0 if $\sum_{i=1}^n (X_i - \bar{X})^2/\sigma_0^2 < C_1$ or $\sum_{i=1}^n (X_i - \bar{X})^2/\sigma_0^2 > C_2$ ” to test the hypothesis $H_0: \sigma = \sigma_0^2$ against the alternative $H_1: \sigma^2 \neq \sigma_0^2$. Use significance level α .
- 12.5.6** Let $X_1, \dots, X_N$ be a random sample from a multinomial distribution with three possible outcomes: O_1, O_2, O_3 and their corresponding probabilities: $p_1, p_2, 1 - p_1 - p_2$. Obtain a UMPU test of a $H_0: p_1 \leq p_2$ against the alternative $H_1: p_1 > p_2$ knowing that among N observation, outcomes O_1, O_2 , and O_3 were observed n_1, n_2 , and $N - n_1 - n_2$ times, respectively.

12.6 GENERALIZED LIKELIHOOD RATIO TESTS

As shown by the Neyman–Pearson lemma, analysis of the likelihood ratio is a good way of searching for test statistics. It turns out that an extension of the likelihood ratio method, originally called the “lambda principle” by Neyman (1950), often leads to tests that perform quite well. The role and importance of these tests, called generalized likelihood ratio (GLR) tests, can be compared with that of maximum likelihood estimators in the estimation problems.

A rather common situation in statistical practice occurs when we are interested in testing hypotheses about a specific parameter, but the population distribution depends also on some other parameters. Equivalently, we may say that $\theta = (\theta_1, \dots, \theta_r)$ is a multidimensional parameter, but we are interested in hypotheses involving one component only. One example here is testing a hypothesis about the population mean μ in the normal distribution when variance σ^2 is also unknown.

The parameter (or parameters) that is not constrained by the null hypothesis is called a *nuisance* parameter. The presence of nuisance parameters causes the null and alternative

hypotheses to be composite. The GLR tests that will be discussed in this section can also be applied to testing in the presence of nuisance parameters.

Suppose that we want to test the null hypothesis $H_0 : \theta \in \Theta_0$ against the alternative $H_1 : \theta \in \Theta \setminus \Theta_0$. In most typical cases both hypotheses H_0 and H_1 are composite, which corresponds to the sets Θ_0 and Θ_1 containing more than one element. In analogy with most powerful tests in the case of a simple null hypothesis and a simple alternative (which are based on the likelihood ratio), we might use the ratio

$$v(\mathbf{X}) = \frac{\sup_{\theta \in \Theta_1} f_n(\mathbf{X}; \theta)}{\sup_{\theta \in \Theta_0} f_n(\mathbf{X}; \theta)}, \quad (12.30)$$

and reject the null hypothesis if the observation $\mathbf{x}$ gives a high value of $v(\mathbf{x})$. The ratio here is based on the analogy with the likelihood ratio tests given by the Neyman–Pearson lemma: if the numerator in (12.30) greatly exceeds the denominator, then $\mathbf{x}$ is good evidence for the alternative hypothesis. On the other hand, small values of $v(\mathbf{x})$ constitute good evidence for the null hypothesis.

An inconvenience with the use of $v(\mathbf{x})$ is that it may be difficult to compute, especially because typically only one of the two suprema in $v(\mathbf{x})$ is attained. Consequently, it is often easier to use the statistic defined as follows:

Definition 12.6.1 The ratio

$$\lambda(\mathbf{X}) = \frac{\sup_{\theta \in \Theta_0} f_n(\mathbf{X}; \theta)}{\sup_{\theta \in \Theta} f_n(\mathbf{X}; \theta)}$$

will be called the *GLR statistic*. □

Under some continuity assumptions, if Θ_0 is a closed set, then the suprema in $\lambda(\mathbf{x})$ are both attained. In particular, if $\hat{\theta}_0 = \hat{\theta}_0(\mathbf{x})$ is the value of the parameter that maximizes $f_n(\mathbf{X}; \theta)$ over the set Θ_0 , then the numerator in $\lambda(\mathbf{x})$ becomes $f_n(\mathbf{X}; \hat{\theta}_0(\mathbf{x}))$. Similarly, the value of θ that gives the maximum of the denominator is simply the MLE of θ , denoted $\hat{\theta} = \hat{\theta}(\mathbf{X})$. Thus, a useful computational formula for $\lambda(\mathbf{X})$ is

$$\lambda(\mathbf{X}) = \frac{\max_{\theta \in \Theta_0} f_n(\mathbf{X}; \theta)}{\max_{\theta \in \Theta} f_n(\mathbf{X}; \theta)} = \frac{f_n(\mathbf{X}; \hat{\theta}_0)}{f_n(\mathbf{X}; \hat{\theta})}.$$

Clearly, $\lambda(\mathbf{X}) \leq 1$, since the denominator, being the maximum over a larger set, is at least as large as the numerator. Since $\lambda(\mathbf{X})$ does not depend on any parameter values, it is a statistic.

To see how $v(\mathbf{X})$ and $\lambda(\mathbf{X})$ are related, observe that

$$\begin{aligned} v(\mathbf{X}) &\leq \frac{\max[\sup_{\theta \in \Theta_1} f_n(\mathbf{X}; \theta), \sup_{\theta \in \Theta_0} f_n(\mathbf{X}; \theta)]}{\sup_{\theta \in \Theta_0} f_n(\mathbf{X}; \theta)} \\ &= \frac{f_n(\mathbf{X}; \hat{\theta})}{\sup_{\theta \in \Theta_0} f_n(\mathbf{X}; \theta)} = \frac{1}{\lambda(\mathbf{X})}; \end{aligned}$$

hence $v(\mathbf{X})\lambda(\mathbf{X}) \leq 1$. Consequently, large values of $v(\mathbf{X})$ are associated with small values of $\lambda(\mathbf{X})$.

It happens sometimes that the distribution of $\lambda(\mathbf{X})$ under H_0 does not depend on any parameters. In such cases, the α -size critical region is obtained from the condition

$$P\{\lambda(\mathbf{X}) \leq k | H_0\} = \alpha. \quad (12.31)$$

The rationale here is that small values of $\lambda(\mathbf{X})$ support the alternative: it means that the maximum of the likelihood over Θ_0 is much smaller than the overall maximum. Hence, it also means that the maximum over Θ is much higher than the maximum over Θ_0 .

It may happen that exact test, using (12.31), is not available. However, it can be shown that if MLE has an asymptotically normal distribution (which is true under very general regularity conditions), then the limiting distribution of $\lambda(\mathbf{X})$ does not involve any parameters.

We have the following theorem, which we provide without proof:

Theorem 12.6.1 *Let $X_1, \dots, X_n$ be a random sample from the $f(x, \theta)$ distribution with $\theta = (\theta_1, \dots, \theta_m)$. If $\theta \in \Theta_0$, then statistic $-2 \log \lambda(\mathbf{X})$ has an asymptotically chi-square distribution with $m - r$ degrees of freedom, where r is the number of components of θ completely specified by the null hypothesis ($r < m$). Thus, the approximate α -size test is*

$$\text{reject } H_0 \text{ if } -2 \log \lambda(\mathbf{X}) \geq \chi_{\alpha, m-r}^2.$$

We will now derive some important GLR tests as examples.

■ EXAMPLE 12.23

Let $\mathbf{X} = (X_1, \dots, X_n)$ be a random sample from the $N(\mu, \sigma^2)$ distribution with μ and σ^2 unknown. For testing $H_0 : \mu = \mu_0, \sigma$ arbitrary, the parameter space Θ is the upper half-plane, while Θ_0 is the ray $\{(\mu, \sigma^2) : \mu = \mu_0\}$. We now have

$$f(\mathbf{X}; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \right\}, \quad (12.32)$$

and the MLE's of μ and σ^2 are

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2. \quad (12.33)$$

After we substitute in (12.32), the denominator in $\lambda(\mathbf{X})$ becomes:

$$\begin{aligned} \max_{(\mu, \sigma^2) \in \Theta} f(\mathbf{X}; \mu, \sigma^2) &= (2\pi)^{-n/2} (\hat{\sigma}^2)^{-n/2} \exp \left\{ -\frac{1}{2\hat{\sigma}^2} \sum_{i=1}^n (X_i - \hat{\mu})^2 \right\} \\ &= \left(2\pi e \sum_{i=1}^n (X_i - \bar{X})^2 / n \right)^{-n/2}. \end{aligned}$$

It remains to find the numerator in $\lambda(\mathbf{X})$, that is,

$$\begin{aligned} \max_{(\mu, \sigma^2) \in \Theta_0} f(\mathbf{X}; \mu, \sigma^2) &= \max_{\sigma^2 > 0} f(\mathbf{X}; \mu_0, \sigma^2) \\ &= \max_{\sigma^2 > 0} (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_0)^2 \right\}. \end{aligned}$$

Here, the maximum is attained at $\hat{\sigma}_0^2 = (1/n) \sum (X_i - \mu_0)^2$ and equals

$$\left(\frac{n}{2\pi e} \right)^{n/2} \left[\sum_{i=1}^n (X_i - \mu_0)^2 \right]^{-n/2}.$$

Consequently,

$$\begin{aligned} \lambda(\mathbf{X}) &= \left(\frac{\sum (X_i - \mu_0)^2}{\sum (X_i - \bar{X})^2} \right)^{-n/2} = \left(\frac{\sum (X_i - \bar{X})^2 + n(\bar{X} - \mu_0)^2}{\sum (X_i - \bar{X})^2} \right)^{-n/2} \\ &= \left(1 + \frac{(\bar{X} - \mu_0)^2}{(1/n) \sum (X_i - \bar{X})^2} \right)^{-n/2} = \left(1 + \frac{1}{n-1} t^2(\mathbf{X}) \right)^{-n/2}, \end{aligned}$$

where

$$t(\mathbf{X}) = \frac{\bar{X} - \mu_0}{\sqrt{(1/n) \sum (X_i - \bar{X})^2}} \sqrt{n-1}$$

has a Student's t distribution with $n-1$ degrees of freedom; see formula (9.21). The inequality $\lambda(\mathbf{X}) \leq k$ is equivalent to the inequality $|t(\mathbf{X})| \geq k^*$.

Thus, the critical region of our test is $|t(\mathbf{X})| \geq t_{\alpha/2, n-1}$. One can show (see Lehmann and Romano, 2005) that this test is UMPU.

■ EXAMPLE 12.24

Consider the GLR test for the hypotheses $H_0 : \mu_1 = \mu_2, \sigma_2^2 = \sigma_1^2$ against the alternative $H_1 : \mu_1 \neq \mu_2, \sigma_2^2 = \sigma_1^2$ (a more general case, solved in the same way, is obtained when we assume that $\sigma_2^2 = \gamma\sigma_1^2$, where γ is a known constant).

The likelihood, letting $\sigma_1^2 = \sigma_2^2 = \sigma^2$, has the form

$$f(\mathbf{x}, \mathbf{y}; \mu_1, \mu_2, \sigma^2) = (2\pi\sigma^2)^{-(m+n)/2} e^{-1/2\sigma^2 [\sum (x_i - \mu_1)^2 + \sum (y_j - \mu_2)^2]}.$$

By maximizing the likelihood over all (μ_1, μ_2, σ^2) and on the subspace $(\mu_1 = \mu_2 = \mu, \sigma^2)$, one can derive the GLR. The exact form of the test is given in Section 12.9. We omit the details of the calculations, which are similar to those in Example 12.23.

■ EXAMPLE 12.25 Paired Observations

A situation deceptively similar to that in Example 12.24 occurs when we have the data obtained by observing the values of some attribute of different elements of the population, observed “before” and “after.” A typical case would be to measure a certain reaction in human subjects before (X_i) and after (Y_i) some treatment. The purpose is to decide whether or not the treatment has an effect. Since we have here the situation of two samples, $X_1, \dots, X_n$ (values “before”) and $Y_1, \dots, Y_n$ (values “after”), we cannot apply the method of Example 12.24 for $m = n$.

Such a procedure would not be correct, since in the present case the values X_i, Y_i are not independent (as observations for the same subject). Under the following assumptions, however, one can use here a one-sample test: The observations need to be such that the differences $U_i = Y_i - X_i$ have the same normal distribution with mean μ and variance σ^2 . Separately, X_i 's and Y_i 's do not need to be normally distributed.

We may wish to test the null hypothesis $H_0 : \mu = \mu_0$ against a one- or two-sided alternative $H_1 : \mu > \mu_0$ or $H_1 : \mu \neq \mu_0, \sigma^2$ arbitrary. In most typical applications, we take $\mu_0 = 0$ (treatment has no effect). The form of the test is the same as in Example 12.24, applied to random variables U_i , and we omit the details.

■ EXAMPLE 12.26

To test the efficiency of sleeping pills, a drug company uses a sample of patients with insomnia. The time (in minutes) until falling asleep is observed for each person. A few days later, the same patients are given a sleeping pill and the time until falling asleep is measured again. Suppose that the data are

Subject	No pill (X_i)	With pill (Y_i)
1	65	45
2	35	5
3	80	61
4	40	31
5	50	20

The proper procedure is to treat the data as paired. The differences $u_i = x_i - y_i$ are then 20, 20, 19, 9, and 30 and we want to test the hypothesis $E(U_i) = 0$ against the alternative $E(U_i) > 0$. Since $\bar{u} = 19.6$ and $s_U^2 = (1/5) \sum (u_i - \bar{u})^2 = 44.24$ we have

$$t = \frac{\bar{u}}{s_U} \sqrt{5 - 1} = 5.89.$$

Compared with the quantiles of the Student's t distribution with 4 degrees of freedom, the result has a p -value below 0.005.

However, if we treat the problem as a two-sample problem (which is *incorrect*), we obtain a different conclusion. The procedure is described in Section 12.9, but it is worthwhile to explain here why an analysis of *the same numerical values* can lead to two different conclusions, depending whether the values result from paired or unpaired data.

In essence, in both cases we are comparing two means, and trying to determine whether their difference is so small that it may be explained by chance variation (null hypothesis) or that it is large enough to be regarded as “significant.” To make such inference, we have to assess the amount of variability, to serve as a base for comparison.

Now the formulas for an estimate of variance are different in case where data are paired and in case they are not. In general, the first estimate gives a lower p -value simply because “a person is typically more similar to himself than to another person.” Thus, quantitatively speaking, the same difference between means can turn out to be significant when compared with a smaller variance given by the formula for paired data, than when compared with higher variance given by the formula for independent samples.

In the present case, we have $\bar{x} = 54$, $s_X = 16.55$, $\bar{y} = 32.4$, and $s_Y = 19.41$, so that

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{5s_X^2 + 5s_Y^2}} \sqrt{\frac{5 + 5 - 2}{\frac{1}{5} + \frac{1}{5}}} = \frac{21.6}{57.04} \sqrt{20} = 1.69,$$

which, for 8 degrees of freedom, is not significant at the 0.05 level.

We complete this section with a discussion of the problem of reaching a decision before the data collection is complete.

■ EXAMPLE 12.27

A research laboratory employs two specialists, Dr. Brown and Dr. Smith. Dr. Brown claims that he invented a certain method that is superior to the currently used method of Dr. Smith. After some debate, it is decided that Dr. Brown's method will be tested. Five experiments are to be run on five consecutive days, starting Monday, and the results $X_1, \dots, X_5$ recorded.²⁷ It is known that X_i 's form a random sample from a

²⁷One may feel that important decisions, like about the superiority of a scientific method, cannot be decided on the basis of only five observations. Of course, if possible, one should use a larger sample size to get a better quality of inference. This, however, does not change the essence of the problem.

normal distribution with an unknown standard deviation. It is also known that for the current method of Dr. Smith, the mean is 10, or perhaps less, so Dr. Brown's method will be declared superior if the mean of X_i 's is higher than 10; that is, if the null hypothesis $H_0 : \mu \leq 10$ (asserting that Dr. Brown's method is no better than Dr. Smith's) is rejected in favor of the alternative $H_1 : \mu > 10$. The significance level, $\alpha = 0.01$, is agreed upon during negotiations.

The observed data for the consecutive 5 weekdays are 14.8, 13.6, 13.9, 10.3, and 11.4. We have here $\bar{x} = 12.8$, and $\sqrt{(1/5) \sum (x_i - \bar{x})^2} = 1.6769$, which gives

$$t = \frac{(12.8 - 10)}{1.6769} \sqrt{4} = 3.339.$$

This value is below the critical value $t_{0.01,4} = 3.747$, so Dr. Brown's method is not declared to be superior at the significance level 0.01.

Dr. Brown, however, is not someone who easily gives up. He notices that if the test were run on Wednesday (after only three observations), we would have $\bar{x} = 14.1$, and $\sqrt{(1/3) \sum (x_i - \bar{x})^2} = 0.51$, hence $t = 11.369$, which exceeds the critical value $t_{0.01,2} = 6.965$, so the p -value would be less than 0.01.

The issues involved here are serious. Generally, when the data are collected sequentially, it happens that the conclusion reached on the basis of all data values differs from a conclusion that is reached on the basis of some initial sequence of data points. Is one then justified in reaching the conclusion that is, for some reason, more convenient (e.g., in favor of one's own preferred hypothesis, and likely to get one an extension of a grant)? In particular, can one discard part of the data?

The answer, of course, is negative, not merely because of the moral issues involved. It is equally important to realize that with modified or discarded data it may be hard, or even impossible, to assess the probabilities involved. To take the example of Dr. Brown, to assess the p -value of the result calculated with the use of the first three data points, one would have to assess the conditional probability, given the conclusion of all five data points, that is,²⁸

$$P\{(X_1, X_2, X_3) \in C_3 | (X_1, \dots, X_5) \notin C_5\},$$

where C_3 and C_5 are critical regions for sample sizes $n = 3$ and $n = 5$.

Quite apart from the moral and computational issues involved, there exists a theory of *sequential* testing of hypotheses. At each new data point, one of the three decisions is made: "accept H_0 ," "accept H_1 ," or "take another observation." The process stops on making either of the first two decisions. The criteria for making these decisions are chosen in such a way that (1) the probabilities of making wrong decisions (type I and type II errors) are bounded by preassigned numbers and (2) the number of observations taken (being a random variable) has finite expectation.

This theory was developed originally by A. Wald. The details can be found in many advanced textbooks on mathematical statistics.

²⁸The analysis here is similar (but more complicated) than the ballot problem, studied in Chapter 3. There we calculated the probability that in the process of counting votes, the losing candidate will lead at least once during the counting. A moment of reflection shows that we have here a very similar situation, except that the "votes" (being random variables with values ± 1) are replaced by observations, that may "favor" one or the other hypothesis in varying degree.

PROBLEMS

- 12.6.1** Suppose that we test the null hypothesis that $\mu = 100$ against the alternative that $\mu \neq 100$. The distribution is normal with its variance unknown. We have just two observations, $X_1 = 105$ and $X_2 = 105 + a$. Find a such that the null hypothesis is rejected at the significance level $\alpha = 0.05$
- 12.6.2** The following data concerning accidents on various types of highways were obtained from the Ohio Department of Transportation, September 1990 (see Al-Ghamdi, 1991):

Highway type	Number of accidents	Annual million vehicle miles	Accident rate
Scenic	3,621	1,021	3.55
Other 2-lane	36,752	11,452	3.21
Multi-lane	20,348	6,920	3.23
Interstate	10,460	9,412	1.11

From the table it appears that the accident rate on interstate highways is significantly lower than on other types of highways, and that on the first three types, the accident rates are essentially the same. Use the likelihood ratio test to verify those claims.

- 12.6.3** The times for the diagnosis and repair of a car with a certain type of problem are assumed to be normally distributed with mean μ and standard deviation $\sigma = 15$ minutes. A mechanic serviced five cars in one day, and it took him a total of 340 minutes. **(i)** Test, at the level $\alpha = 0.05$, hypothesis $H_0 : \mu \leq 60$ against the alternative $H_1 : \mu > 60$. **(ii)** Suppose that you doubt the information that $\sigma = 15$, and decide to test the hypotheses in (i) without assuming anything about σ . If the sum of squares of the five diagnose/repair times is m , how small should m be to reject H_0 at the significance level $\alpha = 0.05$?
- 12.6.4** Let $X_1, \dots, X_n$ be a random sample from the $\text{GAM}(3, \theta)$ distribution. Derive the GLR test for $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$.
- 12.6.5** A random sample of size n was selected from the $\text{EXP}(\theta)$ distribution. Perform the GLR test of $H_0 : \theta = 2$ against the alternative $H_1 : \theta \neq 2$ if the actual observations are 0.57, 0.21, 2.18, 0.85, 1.44. Use $\alpha = 0.05$.
- 12.6.6** Derive the GLR test for $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ based on a random sample $X_1, \dots, X_n$ selected from the $\text{BETA}(1, \theta)$ distribution. Determine an approximate critical value for a size α .
- 12.6.7** Two independent samples $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ were selected from the $\text{EXP}(\theta_1)$ and $\text{EXP}(\theta_2)$ distributions, respectively. **(i)** Derive the GLR test of size 0.05 for $H_0 : \theta_1 = \theta_2$ against $H_1 : \theta_1 \neq \theta_2$. **(ii)** Perform the test at 0.05 significance level based on observations of $X : 0.81, 1.33, 2.10, 0.67, 0.23, 1.85$, and $Y : 0.04, 5.56, 1.92, 0.12, 0.94, 1.56, 2.31, 0.41$.
- 12.6.8** Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be two independent random samples selected from distributions with $P(X = x) = p_1(1 - p_1)^x, x = 0, 1, \dots$ and $P(Y = y) = p_2(1 - p_2)^y, y = 0, 1, \dots$, respectively. Test $H_0 : p_1 = p_2$ against $H_1 : p_1 \neq p_2$ at 0.1 significance level. Assume that $m = 5, n = 10$ and the actual data values are: 1, 0, 3, 5, 1 and 4, 2, 1, 1, 3, 7, 5, 3, 4, 0.

12.6.9 A survey on a proportion of population using a certain product was conducted in four different cities, in each 200 people were interviewed. Test the hypothesis that the proportion of users of the product is the same in each city if the numbers of people using the product were 34, 52, 41, 45, respectively. Use $\alpha = 0.10$.

12.7 CONDITIONAL TESTS

As explained in the previous section, the parameter (or parameters) that are not constrained by the null hypothesis are called *nuisance* parameters. Besides the likelihood ratio tests, for testing in the presence of nuisance parameters one can use conditional tests.

Suppose that the distribution of X depends on a parameter $\theta = (\eta, \tau)$, where η is the parameter tested and τ is the nuisance parameter. Assume that we want to test the null hypothesis that $\eta = \eta_0$ against the alternative that $\eta < \eta_0$. These are in fact composite hypotheses:

$$H_0 : \eta = \eta_0, \tau \text{ arbitrary} \quad \text{against} \quad H_1 : \eta < \eta_0, \tau \text{ arbitrary}.$$

It may happen that there exists a sufficient statistic, T , for τ . In such a case, the conditional distribution of X given $T = t$ does not depend on τ , and it may then happen that one can find a test on a given level α for H_0 (for each t separately).

We will first illustrate such a situation by an example.

■ EXAMPLE 12.28

Let X and Y be independent binomial random variables: $X \sim \text{BIN}(n_1, p_1)$ and $Y \sim \text{BIN}(n_2, p_2)$. We want to test the hypothesis that $p_1 = p_2$ against one- or two-sided alternative. If the null hypothesis is true, then letting $p = p_1 = p_2$, we have

$$\begin{aligned} P\{X = x | X + Y = k\} &= \frac{P\{X = x, X + Y = k\}}{P\{X + Y = k\}} \\ &= \frac{P\{X = x\}P\{Y = k - x\}}{P\{X + Y = k\}} \\ &= \frac{\binom{n_1}{x} p^x q^{n_1-x} \binom{n_2}{k-x} p^{k-x} q^{n_2-k+x}}{\binom{n_1+n_2}{k} p^k q^{n_1+n_2-k}} \\ &= \frac{\binom{n_1}{x} \binom{n_2}{k-x}}{\binom{n_1+n_2}{k}}, \end{aligned} \tag{12.34}$$

which is independent of the nuisance parameter p .

The question is which values of X (given $X + Y = k$) constitute the strongest evidence against the null hypothesis, and how should one assess the p -value of the result. Here the argument is as follows: We can expect X/n_1 to be close to p_1 and $Y/n_2 = (k - X)/n_2$ to be close to p_2 . So, if H_0 is true, we can expect $X/n_1 \approx k/n_2 - X/n_2$. For the two-sided alternative, the “worst” cases are where X is close to 0 or close to k so that the critical region comprise two tails of the hypergeometric distribution (12.34).

Letting

$$u(j) = u(j; n_1, n_2, k) = \frac{\binom{n_1}{j} \binom{n_2}{k-j}}{\binom{n_1+n_2}{k}},$$

we may define the p -value of the result $X = x$ for two-sided test as

$$2\min \left\{ \sum_{j \leq x} u(j), \sum_{j \geq x} u(j) \right\}. \quad (12.35)$$

This is the Fisher's exact test, which will be discussed in more detail in Chapter 15.

■ EXAMPLE 12.29

Let X and Y be independent observations of two Poisson random variables, with parameters λ_1 and λ_2 , respectively. We want to test the hypothesis $H_0 : \lambda_2 = \delta\lambda_1$ against the alternative $H_1 : \lambda_2 > \delta\lambda_1$, where $\delta > 0$ is some fixed constant. For instance, if $\delta = 1$, we have the hypothesis of equality of parameters in two Poisson distributions.

The joint distribution of (X, Y) is here

$$\begin{aligned} f(x, y; \lambda_1, \lambda_2) &= \frac{\lambda_1^x}{x!} \times \frac{\lambda_2^y}{y!} e^{-(\lambda_1 + \lambda_2)} \\ &= \frac{1}{x!y!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^x \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^y (\lambda_1 + \lambda_2)^{x+y} e^{-(\lambda_1 + \lambda_2)}. \end{aligned}$$

This suggests reparametrization with $\eta = \lambda_2/\lambda_1$ and $\tau = \lambda_1 + \lambda_2$, leading to the joint probability function

$$f(x, y; \eta, \tau) = \binom{x+y}{x} \left(\frac{1}{1+\eta} \right)^x \left(\frac{\eta}{1+\eta} \right)^y \frac{\tau^{x+y}}{(x+y)!} e^{-\tau}. \quad (12.36)$$

It is now clear that $T = X + Y$ is a sufficient statistic for τ , and that given $T = t$, the random variable X has the conditional distribution that is binomial with parameters t and $1/(1 + \eta)$. Under the null hypothesis, we have $\eta = \delta$, while under the alternative hypothesis, $\eta > \delta$. Given $T = t$, we may therefore test the null hypothesis $H_0 : p = 1/(1 + \delta)$ against the alternative $H_1 : p < 1/(1 + \delta)$, observing the random variable X with the $\text{BIN}(t, p)$ distribution.

■ EXAMPLE 12.30

Suppose that traffic engineers suggest a certain change of a traffic light sequence to reduce the number of accidents at some type of intersections. Two intersections, far apart to ensure independence but otherwise identical in all aspect (traffic intensity, road condition, etc.) are to be tested. Over a certain time, the number X of accidents at the intersection with the new traffic light pattern is 7. During the same time, the number of accidents at the intersection with a previous traffic light pattern is $Y = 13$. At the significance level $\alpha = 0.05$, does this indicate that the new traffic light pattern decreases the probability of an accident?

We want to test the hypothesis that $\lambda_1 = \lambda_2$ against the alternative $\lambda_1 < \lambda_2$ (so that $\delta = 1$ and $1/(1 + \delta) = 1/2$). Since $t = x + y = 7 + 13 = 20$ and $X \sim \text{BIN}(20, 1/2)$, the p value is

$$P\{X \leq 7\} = \sum_{j=0}^7 \binom{20}{j} \left(\frac{1}{2} \right)^{20} = 0.1310.$$

The answer is clearly negative. If the new traffic lights pattern does not affect the probability of accident at all, the outcome as obtained (13 against 7), or more extreme, has more than 13% chance of occurring.

The scheme above is an application of the following theorem:

Theorem 12.7.1 Suppose that the random sample $\mathbf{X} = (X_1, \dots, X_n)$ has a joint distribution of the form

$$f_n(\mathbf{x}, \eta, \tau_1, \dots, \tau_r) = a(\eta, \tau_1, \dots, \tau_r) b(\mathbf{x}) \exp \left\{ \eta u(\mathbf{x}) + \sum_{j=1}^r \tau_j v_j(\mathbf{x}) \right\} \quad (12.37)$$

for some functions $a, b, u, v_1, \dots, v_r$. Let $U = u(\mathbf{X})$, and let $\mathbf{V} = (V_1, \dots, V_r)$, where $V_j = v_j(\mathbf{X})$ for $j = 1, \dots, r$. Then for each η , variables $V_1, \dots, V_r$ are jointly sufficient for $(\tau_1, \dots, \tau_r)$, and the conditional distribution of U given $V(\mathbf{x}) = \mathbf{v}$ depends on η but not on $(\tau_1, \dots, \tau_r)$. A test with size α for testing $H_0: \eta \leq \eta_0$ against $H_1: \eta > \eta_0$ is obtained by rejecting H_0 if $U(\mathbf{x}) \geq q(\mathbf{v})$, where $P\{U \geq q(\mathbf{v}) | \mathbf{v}\} = \alpha$ for $\eta = \eta_0$.

For testing $H_0: \eta \geq \eta_0$ against $H_1: \eta < \eta_0$ the directions of inequalities will be reversed; a two-tailed test is to be used in case of two-sided hypothesis $H_0: \eta = \eta_0$ against $H_1: \eta \neq \eta_0$.

Lehmann and Romano (2005) show that under some regularity conditions these tests are UMPU.

■ EXAMPLE 12.31

To see how the situation of Example 12.29 falls under the scheme of Theorem 12.7.1, observe that the distribution of $\mathbf{X} = (X_1, X_2)$ can be reduced to the form (12.37) with $r = 1$, $U(\mathbf{X}) = X_1$, $V(\mathbf{X}) = X_1 + X_2$. This is obtained as follows: Take $s = \lambda_2/\lambda_1$, $t = \lambda_1 + \lambda_2$; hence $\lambda_1 = t/(1+s)$, $\lambda_2 = st/(1+s)$. The distribution

$$f(x_1, x_2; \lambda_1, \lambda_2) = \frac{1}{x_1!} \lambda_1^{x_1} e^{-\lambda_1} \times \frac{1}{x_2!} \lambda_2^{x_2} e^{-\lambda_2}$$

reduces, after some algebra, to the form

$$f = \binom{V(\mathbf{x})}{U(\mathbf{x})} \frac{1}{V(\mathbf{x})!} e^{-t} \exp \left\{ U(\mathbf{x}) \ln \frac{1}{s} + V(\mathbf{x}) \left[\log t + \log \frac{s}{1+s} \right] \right\}.$$

Introducing new parameters $\eta = \log(1/s) = \log(\lambda_1/\lambda_2)$ and $\tau = \log t + \log s/(1+s) = \log \lambda_2$, and then expressing t in terms of τ and η , we obtain a density in the form (12.37).

PROBLEMS

12.7.1 Suppose that in a group of 10 randomly sampled Democrats only 2 favor a certain issue, whereas in a sample of 12 Republicans the same issue is favored by 5 persons. At the level $\alpha = 0.05$, does this result indicate that the fractions p_D and p_R of Democrats and Republicans favoring the issue in question are different? Find the p -value by carrying out the two-sided Fisher's test.

12.7.2 A company A that produces batteries claims that their product lasts “at least 50% longer” than the batteries produced by company B . To test the claim, batteries A and B are used one after another in two analogue devices. That is, one device has a battery A installed and is left running until the battery becomes dead. It is then immediately replaced by another battery A , and so on. The second device runs parallel on batteries B .

Assume that the lifetimes of the batteries are exponential random variables, with densities $\lambda_A e^{-\lambda_A t}$ and $\lambda_B e^{-\lambda_B t}$. Suppose that in some time (e.g., a week) batteries A had to be replaced 5 times, whereas batteries B had to be replaced 9 times. Test the advertising claim by determining the p -value of the result. (*Hint*: If interarrival times are exponential, the process is Poisson.)

12.8 TESTS AND CONFIDENCE INTERVALS

In this section, we will briefly explain how the theory of testing statistical hypotheses is related to the theory of confidence intervals, discussed in Chapter 11. This connection was noticed by Neyman (who laid the foundations to both theories) as early as in 1938. To simplify the presentation, assume that $X_1, \dots, X_n$ is a random sample from a distribution $f(x; \theta)$, and that θ is a one-dimensional parameter.

A confidence interval (with confidence level $1 - \alpha$) is a random interval $[L, U] = [L(\mathbf{X}), U(\mathbf{X})]$ such that for every $\theta \in \Theta$,

$$P_\theta\{L(\mathbf{X}) \leq \theta \leq U(\mathbf{X})\} = 1 - \alpha. \quad (12.38)$$

Suppose that we want to test hypothesis $H_0 : \theta = \theta_0$ against the alternative $H_1 : \theta \neq \theta_0$. The equivalence of tests and confidence intervals is based on the fact that if we can construct confidence interval (12.38), then we can also construct an α -level test of H_0 against H_1 , and conversely: given a testing procedure of level α , we can construct a confidence interval.

Indeed, condition (12.38) for $\theta = \theta_0$ allows us to define the set

$$A = \{\mathbf{x} = (x_1, \dots, x_n) : L(\mathbf{x}) \leq \theta_0 \leq U(\mathbf{x})\}.$$

Clearly, if we take the set A as the acceptance region of H_0 (equivalently, we let $C = A^c$ to be the critical region for H_0), we obtain a test with level α :

$$P\{H_0 \text{ is rejected} | H_0 \text{ is true}\} = P\{\theta_0 \notin [L(\mathbf{X}), U(\mathbf{X})] | \theta = \theta_0\} = \alpha.$$

Conversely, suppose that for every θ' we can construct an α -level test of the hypothesis $H_0 : \theta = \theta'_0$. This means that for every θ' we have a critical region $C_{\theta'}$ such that

$$P\{\mathbf{X} \notin C_{\theta'} | \theta = \theta'\} = 1 - \alpha. \quad (12.39)$$

Define now, for every $\mathbf{x}$,

$$B(\mathbf{x}) = \{\theta' : \mathbf{x} \notin C_{\theta'}\}. \quad (12.40)$$

When $\mathbf{x}$ is the observed value of random vector $\mathbf{X}$, we obtain a random set $B(\mathbf{X})$. From (12.39) and (12.40) it follows that $P_\theta\{\theta \in B(\mathbf{X})\} = 1 - \alpha$, which means that $B(\mathbf{X})$ is a confidence set for θ with confidence level $1 - \alpha$.

Except for the fact that the second part of the argument provides us with confidence sets (not necessarily intervals), the argument shows that the two theories are essentially equivalent, at least in case of one-dimensional parameters.

The main results of the theory combine optimality properties of tests and confidence intervals. For instance, confidence intervals (sets) associated with UMP tests have the property of being the shortest possible (uniformly most accurate, or UMA confidence intervals). It is worth mentioning here that the theory extends also to the case of testing in presence of nuisance parameters, but we will not go into the details here.

12.9 REVIEW OF TESTS FOR NORMAL DISTRIBUTIONS

We now review the major testing procedures for parameters of normal distributions. This section is intended as a convenient reference for users rather than an exposition of new concepts or results. If the derivations were given in the other sections, we refer to them. In other cases, we omit the derivations, specifying only the properties of the tests, possibly with indications of the proofs.

In all tests below, α is the significance level and $1 - \beta$ is the power, z_p is the upper p th quantile of standard normal random variable Z , so that

$$P\{Z > z_p\} = 1 - \Phi(z_p) = p.$$

Similarly, $t_{p,\nu}$ is the upper p th quantile of the Student's t distribution with ν degrees of freedom, $\chi^2_{p,\nu}$ is the upper p th quantile of the chi-square distribution with ν degrees of freedom, and F_{p,ν_1,ν_2} denotes an upper p th quantile of the F distribution with ν_1 and ν_2 degrees of freedom.

One-Sample Procedures

Let $X_1, \dots, X_n$ be a random sample from distribution $N(\mu, \sigma^2)$.

Hypotheses About the Mean, Variance Known

The sufficient statistic is $\sum_{i=1}^n X_i$, hence also $\bar{X}$.

- Hypotheses:

One-sided alternatives:

(i) $H_0 : \mu = \mu_0$ (or $\mu \leq \mu_0$) vs. $H_1 : \mu > \mu_0$.

(ii) $H_0 : \mu = \mu_0$ (or $\mu \geq \mu_0$) vs. $H_1 : \mu < \mu_0$.

Two-sided alternatives:

(iii) $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$.

(iv) $H_0 : \mu_1 \leq \mu \leq \mu_2$ vs. $H_1 : \mu < \mu_1$ or $\mu > \mu_2$.

em Other cases:

(v) $H_0 : \mu \leq \mu_1$ or $\mu \geq \mu_2$ vs. $H_1 : \mu_1 < \mu < \mu_2$.

- Test statistics:

(i)–(iii) $T_1 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$.

(iv), (v) $T_2 = \frac{\bar{X} - \mu^*}{\sigma/\sqrt{n}}$, where $\mu^* = \frac{\mu_1 + \mu_2}{2}$.

- Corresponding critical regions, reject H_0 if:

(i) $T_1 \geq z_\alpha$, (ii) $T_1 \leq z_{1-\alpha}$, (iii) $|T_1| \geq z_{\alpha/2}$, (iv) $|T_2| \geq k_\alpha$,

(v) $|T_2| \leq v_\alpha$, where k_α and v_α are determined from

$$\Phi\left(\frac{\Delta}{\sigma/\sqrt{n}} + k_\alpha\right) - \Phi\left(\frac{\Delta}{\sigma/\sqrt{n}} - k_\alpha\right) = 1 - \alpha$$

and

$$\Phi\left(\frac{\Delta}{\sigma/\sqrt{n}} + v_\alpha\right) - \Phi\left(\frac{\Delta}{\sigma/\sqrt{n}} - v_\alpha\right) = \alpha,$$

with $\Delta = (\mu_2 - \mu_1)/2$. Tests (i), (ii), and (v) are UMP tests; (iii) and (iv) are UMPU tests.

- Power:

(i) $\pi(\mu) = 1 - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_\alpha\right)$

- (ii) $\pi(\mu) = 1 - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_\alpha\right)$
- (iii) $\pi(\mu) = 1 - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) + \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$
- (iv) $\pi(\mu) = 1 - \Phi\left(\frac{\mu^* - \mu}{\sigma/\sqrt{n}} + k_\alpha\right) + \Phi\left(\frac{\mu^* - \mu}{\sigma/\sqrt{n}} - k_\alpha\right)$
- (v) $\pi(\mu) = 1 - \Phi\left(\frac{\mu^* - \mu}{\sigma/\sqrt{n}} + v_\alpha\right) + \Phi\left(\frac{\mu^* - \mu}{\sigma/\sqrt{n}} - v_\alpha\right).$

• Sample size determination:

Wanted: sample size n giving power at least $1 - \beta$ at μ

(i), (ii) $n \geq \frac{(z_{1-\alpha} + z_{1-\beta})^2}{(\mu - \mu_0)^2} \sigma^2$

(iii) Solve numerically for n ,

$$\Phi\left(\frac{\mu_0 - \mu}{\sigma} \sqrt{n} + z_{1-\alpha/2}\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma} \sqrt{n} - z_{1-\alpha/2}\right) = \beta.$$

(iv) Solve numerically for n ,

$$\Phi\left(\frac{\mu^* - \mu}{\sigma} \sqrt{n} + k_\alpha\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma} \sqrt{n} - k_\alpha\right) = \beta.$$

(v) Solve numerically for n ,

$$\Phi\left(\frac{\mu^* - \mu}{\sigma} \sqrt{n} + v_\alpha\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma} \sqrt{n} - v_\alpha\right) = \beta.$$

■ **EXAMPLE 12.32 Generic Problem**

All philogaps presently on the market have an average concentration of muzzzz of at least 3.7 mg per philogap. A company claims to have discovered a new method of production that will decrease the average muzzzz content to the level below 3.7 mg. To test this claim, the muzzzz content of 15 philogaps of this company are analyzed, and their average muzzzz content is found to be 3.52 mg.

It is known that the standard deviation of the muzzzz content in a philogap does not depend on the production process and equals 0.35 mg. It is also known that the muzzzz content is normally distributed. At the significance level $\alpha = 0.01$, does this finding indicate that the new production process decreases the concentration of muzzzz in philogaps?

Remark You are probably curious about what are philogaps and what is muzzzz (spelled with triple z). These words, to our best knowledge, mean nothing. If you so wish, substitute “objects” and “attribute A ,” or “cars” and “miles per gallon,” “beer” and “alcohol content,” “oranges” and “sugar content of juice,” and so on, and change numbers and possibly, inequality directions accordingly.

We set up this problem as being of type (ii): The null hypothesis is $H_0 : \mu \geq 3.7$ and the alternative is $H_1 : \mu < 3.7$. The value of the test statistic T_1 is $t = [(3.52 - 3.7)/0.35]\sqrt{15} = -1.99$. The 1% quantile of the standard normal distribution is -2.33 , so the null hypothesis is not rejected. The average 3.52 or less in a sample of 15 is more likely than 0.01, if the mean is in fact 3.7. Therefore, the null hypothesis cannot be rejected at the significance level 0.01. The p -value is equal to $P(Z < -1.99) = 0.0233$, or about 2.3%.

Suppose that we want not only the 1% level of significance but also at least 95% chance of detecting the improvement of the average muzzz content in philogaps by 0.15 mg. In statistical terms, this means that we want the power to be at least 0.95 at $\mu = 3.7 - 0.15 = 3.55$. Then, we need to take the sample of at least

$$\frac{(z_{0.01} + z_{0.05})^2}{(3.7 - 3.55)^2} (0.35)^2 = \frac{(2.33 + 1.96)^2}{(0.15)^2} (0.35)^2 = 100.2,$$

meaning that n should be at least 101.

■ EXAMPLE 12.33

A food-packing company purchased a new machine to fill plastic containers with sour cream. The nominal weight, as listed on the container, is 8 oz. The dial on the machine can be set on average weight $\dots, 7.98, 8.00, 8.02, 8.04, \dots$ oz. When it is set on 8.00 oz (say), it puts into successive containers the amounts $X_1, X_2, \dots$, which are normally distributed with some mean μ and standard deviation (the same for all settings of the dial) $\sigma = 0.005$ oz. This standard deviation reflects the unavoidable container-to-container variability of the amounts of sour cream about their mean μ . Naturally, it is impossible to build a machine that gives the average μ *exactly* equal to the setting on the dial, be it 7.995, 8.002, and so on.

A consumer protection agency may disregard the instances where the variability will occasionally lead to a container with more than the nominal amount of sour cream. However, it might strongly object if the average μ is even slightly less than 8 oz, as this constitutes a systematic theft from society. On the other hand, if μ exceeds the nominal weight even slightly, it may in time constitute a sizable free gift of the company to society.

The company decides that it will risk getting into trouble with the consumer protection agency, or absorb the loss, if the mean μ satisfies the inequality $7.995 \leq \mu \leq 8.015$ but wants to avoid both lower μ and higher μ . Careful measurements of 50 containers with sour cream give the average $\bar{x} = 8.017$.

We are now in situation (iv) where the null hypothesis states that μ lies between some bounds, and we may proceed as follows: We have here $\mu_1 = 7.995$ and $\mu_2 = 8.015$, so $\mu^* = 8.005$ oz and $\Delta = 0.01$. Consequently, the observed value of T_2 is $t_2 = [(8.017 - 8.005)/0.005]\sqrt{50} = 16.97$. On the other hand, $(\Delta/\sigma)\sqrt{n} = 14.142$, and the equation

$$\Phi(14.142 + k_\alpha) - \Phi(14.142 - k_\alpha) = 1 - \alpha$$

reduces to $\Phi(14.142 - k_\alpha) = \alpha$. For $\alpha = 0.01$ we must have $14.142 - k_\alpha = -2.33$; hence, $k_\alpha = 16.472$. The observed value 16.97 exceeds k_α , and this indicates that the null hypothesis should be rejected at the level 0.01.

The conclusion is that $\mu < 7.995$ or $\mu > 8.015$ despite the fact that the result $\bar{x} = 8.017$ suggests that the second of the two inequalities holds. The point is that the null hypothesis states that $7.995 \leq \mu \leq 8.015$, and its rejection is logically equivalent to the pair of inequalities.

To conclude that $\mu > 8.015$, we should test the null hypothesis $\mu \leq 8.015$. The test statistic is then

$$t_1 = \frac{8.017 - 8.015}{0.005} \sqrt{50} = 2.82,$$

with a corresponding p -value of about 0.0025.

Hypotheses About the Mean, Variance Unknown

The jointly sufficient statistics are $\sum X_i$ and $\sum X_i^2$, or equivalently $\bar{X}$ and $\sum (X_i - \bar{X})^2$.

- Hypotheses:

One-sided alternative:

(i) $H_0 : \mu = \mu_0, \sigma > 0$ (or $H_0 : \mu \leq \mu_0, \sigma > 0$) vs. $H_1 : \mu > \mu_0, \sigma > 0$.

(ii) $H_0 : \mu = \mu_0, \sigma > 0$ (or $H_0 : \mu \geq \mu_0, \sigma > 0$) vs. $H_1 : \mu < \mu_0, \sigma > 0$. *Two-sided alternative:*

(iii) $H_0 : \mu = \mu_0, \sigma > 0$ vs. $H_1 : \mu \neq \mu_0, \sigma > 0$.

- Test statistic:

$$t = \frac{\bar{X} - \mu_0}{\sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}} \sqrt{n-1}.$$

Remark Many tests use the notation $t = [(\bar{X} - \mu_0)/S]\sqrt{n-1}$ or $t = [(\bar{X} - \mu_0)/S]\sqrt{n}$. These may be confusing, since one has to bear in mind that in the first case S^2 is the MLE of σ^2 ; that is, $S^2 = (1/n) \sum (X_i - \bar{X})^2$. In the second case, S^2 is the unbiased estimator of variance: $S^2 = [1/(n-1)] \sum (X_i - \bar{X})^2$. The notation for S^2 is *not* standardized across statistical textbooks and papers.

- Corresponding critical regions, reject H_0 if:

(i) $t \geq t_{\alpha, n-1}$.

(ii) $t \leq -t_{\alpha, n-1}$.

(iii) $|t| \geq t_{\alpha/2, n-1}$.

For each of these tests the power is the function π depending on a two-dimensional variable (μ, σ^2) . We have $\pi(\mu_0, \sigma^2) = \alpha$ for all σ^2 . However, for $\mu \neq \mu_0$, the values $\pi(\mu, \sigma^2)$ depend on (unknown) σ^2 , and are given by the noncentral Student distribution. Therefore, the sample size determination requires some additional information about σ^2 (a two-stage procedure, etc.). All three tests are UMPU²⁹ tests.

■ EXAMPLE 12.34

A certain make of cars is advertised as attaining gas mileage of at least 32 miles per gallon. Twelve independent tests gave the results 33, 28, 31, 28, 26, 30, 31, 28, 27, 33, 35, 29 miles per gallon. What can one say about the advertisements in light of these data?

Let us make the assumption that the observed mileages are normally distributed. We may then set the problem of evaluation of the advertising claim as that of testing the hypothesis $H_0 : \mu \geq 32$ (claim is true) against the alternative $H_1 : \mu < 32$ (claim is false). We have here $n = 12$, $\sum x_i = 359$, $\sum x_i^2 = 10,823$. Thus $\bar{x} = 29.92$, $\sum (x_i - \bar{x})^2 = 82.92$, so $t = -2.62$. Since the critical values for 11 degrees of freedom are $t_{0.025, 11} = 2.201$ and $t_{0.01, 11} = 2.718$, the result is not significant at the 0.01 level but significant at the 0.025 level. In other words, the p -value is between 0.01 and 0.025.

Hypotheses About the Variance, Mean Known

The sufficient statistic is $\sum (X_i - \mu)^2$. The sample size can be 1.

- Hypotheses:

One-sided alternative:

²⁹As mentioned before, two-sided UMP tests may not exist. Nevertheless, they can often be found in some restricted classes of tests, for example, in the class of unbiased tests.

- (i) $H_0 : \sigma^2 = \sigma_0^2$ (or $H_0 : \sigma^2 \leq \sigma_0^2$ vs. $H_1 : \sigma^2 > \sigma_0^2$).
- (ii) $H_0 : \sigma^2 = \sigma_0^2$ (or $H_0 : \sigma^2 \geq \sigma_0^2$ vs. $H_1 : \sigma^2 < \sigma_0^2$).

Two-sided alternative:

- (iii) $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 \neq \sigma_0^2$.

- Test statistic:

$$U = \sum_{i=1}^n (X_i - \mu)^2 / \sigma_0^2.$$

- Corresponding critical regions, reject H_0 if:

- (i) $U < \chi_{\alpha,n}^2$.
- (ii) $U > \chi_{1-\alpha,n}^2$.
- (iii) $U < \chi_{1-\alpha/2,n}^2$ or $U > \chi_{\alpha/2,n}^2$.

- Power:

(i)

$$\begin{aligned} \pi(\sigma^2) &= 1 - \beta(\sigma^2) = P_{\sigma^2} \{U \leq \chi_{\alpha,n}^2\} = P_{\sigma^2} \left\{ \frac{\sum (X_i - \mu)^2}{\sigma_0^2} \leq \chi_{\alpha,n}^2 \right\} \\ &= P_{\sigma^2} \left\{ \frac{\sum (X_i - \mu)^2}{\sigma^2} \leq \sigma_0^2 \frac{\chi_{\alpha,n}^2}{\sigma^2} \right\} = F_n \left(\sigma_0^2 \frac{\chi_{\alpha,n}^2}{\sigma^2} \right). \end{aligned}$$

Calculations for remaining cases are similar.

Tests (i) and (ii) are UMP. Test (iii) is asymptotically (for $n \rightarrow \infty$) UMPU. A UMPU test in case (iii) is obtained if the thresholds are equal to the endpoints of the shortest $(1 - \alpha)$ -level confidence interval for σ^2 (see Section 11.7).

Hypotheses About the Variance, Mean Unknown

Jointly sufficient statistics are $\bar{X}$ and $\sum (X_i - \bar{X})^2$. Tests require $n \geq 2$.

- Hypotheses:

One-sided alternative:

- (i) $H_0 : \sigma^2 = \sigma_0^2$ (or $H_0 : \sigma^2 \leq \sigma_0^2$) vs. $H_1 : \sigma^2 > \sigma_0^2$; $\mu \in R$.
- (ii) $H_0 : \sigma^2 = \sigma_0^2$ (or $H_0 : \sigma^2 \geq \sigma_0^2$) vs. $H_1 : \sigma^2 < \sigma_0^2$; $\mu \in R$.

Two-sided alternative:

- (iii) $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 \neq \sigma_0^2$; $\mu \in R$.

- Test statistic:

$$V = \sum_{i=1}^n (X_i - \bar{X})^2 / \sigma_0^2.$$

- Corresponding critical regions, reject H_0 if:

- (i) $V > \chi_{\alpha,n-1}^2$.
- (ii) $V < \chi_{1-\alpha,n-1}^2$.
- (iii) $V < \chi_{1-\alpha/2,n-1}^2$ or $V > \chi_{\alpha/2,n-1}^2$.

- Power:

$\pi(\mu, \sigma^2) = P_{\mu, \sigma^2} \{H_0 \text{ is rejected}\}$ is obtained in the same way as in the case above.

Test (i) is unbiased. Tests (ii) and (iii) are not UMPU.

The following example was taken from Larsen and Marx (1986):

■ EXAMPLE 12.35

The A above middle C is the note given to an orchestra, usually by the oboe, for tuning purposes. Its pitch is defined to be the sound of a tuning fork vibrating at 440 hertz (Hz). No tuning fork, of course, will always vibrate at *exactly* 440 Hz; rather, the pitch, Y , is a random variable. Suppose that Y is normally distributed with $\mu = 440$ Hz and variance σ^2 (here the parameter σ^2 is a measure of quality of the tuning fork). With the standard manufacturing process, $\sigma^2 = 1.1$. A new production technique has just been suggested, however, and its proponents claim it will yield values of σ^2 significantly less than 1.1. To test the claim, six tuning forks are made according to the new procedure. The resulting vibration frequencies are 440.8, 440.3, 439.2, 439.8, 440.6, and 441.1 Hz.

We will test the hypothesis $H_0 : \sigma^2 = 1.1$ against the alternative $H_1 : \sigma^2 < 1.1$, at the significance level $\alpha = 0.05$. First, we accept as the fact that the new production process gives the mean $\mu = 440$. Then $u = \sum (x_i - 440)^2 / 1.1 = 1.62$. The value $\chi_{0.05,6}^2 = 1.635$, so there is evidence in the data (at $\alpha = 0.05$ significance level) that in the new production process the variance is less than 1.1.

Suppose that we have some doubts about whether or not $\mu = 440$. We can then use the method for unknown μ . Now $v = 2.219$, $\chi_{0.05,5}^2 = 1.145$, and the conclusion is now different—the claim is not supported by the evidence in the data.

Two-Sample Procedures

Two independent random samples $X_1, \dots, X_m$ and $Y_1, \dots, Y_m$ are selected from distributions $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively.

Hypotheses About the Means, Variances Known

Jointly sufficient statistics are $\sum_{i=1}^m X_i$ and $\sum_{i=1}^n Y_i$, or $\bar{X}$, and $\bar{Y}$.

- Hypotheses:

One-sided alternative:

(i) $H_0 : \mu_1 = \mu_2$ (or $H_0 : \mu_1 \leq \mu_2$) vs. $H_1 : \mu_1 > \mu_2$.

Two-sided alternative:

(ii) $H_0 : \mu_1 = \mu_2$ vs. $H_1 : \mu_1 \neq \mu_2$.

Remark The opposite inequality in (i) reduces to changing the role of X_i 's and Y_j 's. The apparently more general hypotheses of the form $\mu_1 = \mu_2 - \Delta$, reduce to the ones above by subtracting constant Δ from all Y_j 's.

- Test statistic:

$$U = \frac{\bar{X} - \bar{Y}}{\sqrt{\sigma_1^2/m + \sigma_2^2/n}}$$

- Corresponding critical regions, reject H_0 if:

(i) $U \geq z_\alpha$.

(ii) $|U| \geq z_{\alpha/2}$.

- Power:

(i) $\pi(\mu_1, \mu_2) = 1 - \beta(\Delta) = 1 - \Phi\left(z_\alpha + \Delta / \sqrt{\sigma_1^2/m + \sigma_2^2/n}\right)$.

$$(ii) \pi(\mu_1, \mu_2) = 1 - \beta(\Delta) = 1 - \Phi\left(z_{\alpha/2} + \Delta/\sqrt{\sigma_1^2/m + \sigma_2^2/n}\right) \\ + \Phi\left(-z_{\alpha/2} + \Delta/\sqrt{\sigma_1^2/m + \sigma_2^2/n}\right),$$

where $\mu_2 - \mu_1 = \Delta$.

Tests (i) is UMP; test (ii) is UMPU.

Hypotheses About the Means, Variances Unknown, but $\sigma_2^2 = \gamma\sigma_1^2$

Jointly sufficient statistics are $\bar{X}$, $\bar{Y}$, and $\sum (X_i - \bar{X})^2 + \frac{1}{\gamma} \sum (Y_j - \bar{Y})^2$.

- Hypotheses:

One-sided alternative:

$$(i) H_0 : \mu_1 = \mu_2 \text{ (or } H_0 : \mu_1 \leq \mu_2) \text{ vs. } H_1 : \mu_1 > \mu_2; \quad \sigma_1^2 > 0.$$

Two-sided alternative:

$$(ii) H_0 : \mu_1 = \mu_2, \text{ vs. } H_1 : \mu_1 \neq \mu_2; \quad \sigma_1^2 > 0.$$

Remark The opposite inequality in (i) reduces to changing the role of X_i 's and Y_j 's. The apparently more general hypotheses of the form $\mu_1 = \mu_2 + \Delta$, reduce to the ones above by adding a constant Δ to all Y_j 's.

- Test statistic:

$$U = \frac{\bar{X} - \bar{Y}}{\sqrt{\sum (X_i - \bar{X})^2 + \frac{1}{\gamma} \sum (Y_j - \bar{Y})^2}} \sqrt{\frac{m+n-2}{1/m + \gamma/n}}.$$

Remark $\gamma = 1$ if $\sigma_1^2 = \sigma_2^2$.

- Corresponding critical regions, reject H_0 if:

$$(i) U > t_{\alpha, m+n-2}.$$

$$(ii) |U| > t_{\alpha/2, m+n-2}.$$

- Power depends on μ_1, μ_2, σ_1^2 , and it is expressed through noncentral Student's t distribution. Tests (i) and (ii) are UMPU.

Hypotheses About the Variances, Means Unknown

Jointly sufficient statistics are $\bar{X}, \bar{Y}, \sum (X_i - \bar{X})^2$, and $\sum (Y_j - \bar{Y})^2$.

- Hypotheses:

One-sided alternative:

$$(i) H_0 : \sigma_2^2 = \gamma\sigma_1^2 \text{ (or } H_0 : \sigma_2^2 \leq \gamma\sigma_1^2) \text{ vs. } H_1 : \sigma_2^2 > \gamma\sigma_1^2; \quad \mu_1 \in R, \mu_2 \in R.$$

Two-sided alternative:

$$(ii) H_0 : \sigma_2^2 = \gamma\sigma_1^2 \text{ vs. } H_1 : \sigma_2^2 \neq \gamma\sigma_1^2; \quad \mu_1 \in R, \mu_2 \in R.$$

Remark $\gamma = 1$ if the hypothesis asserts equality of variances.

- Test statistic:

$$F = \frac{\sum (Y_i - \bar{Y})^2 / \gamma(n-1)}{\sum (X_j - \bar{X})^2 / (m-1)}.$$

- Corresponding critical regions, reject H_0 if:

$$(i) F \geq F_{\alpha, n-1, m-1}.$$

$$(ii) F \leq F_{1-\alpha/2, n-1, m-1} = 1/F_{\alpha/2, m-1, n-1} \text{ or } F \geq F_{\alpha/2, n-1, m-1}.$$

Tests (i) and (ii) are UMPU.

Hypotheses About the Variances; One or Both Means Known

Jointly sufficient statistics are $\bar{X}, \bar{Y}, \sum (X_i - \bar{X})^2, \sum (Y_j - \bar{Y})^2$. Whenever a mean is known, it replaces the sample average in ratio F , and the divisor (and number of degrees of freedom) changes into m (respectively, n). For instance, suppose that both μ_1 and μ_2 are known. Then we test $H_0 : \sigma_2^2 = \sigma_1^2$ vs. $H_1 : \sigma_2^2 > \sigma_1^2$. The testing variable is

$$F = \frac{\sum (Y_j - \mu_2)^2/n}{\sum (X_i - \mu_1)^2/m},$$

and the null hypothesis is rejected if $F \geq F_{\alpha, n, m}$.

Large Sample Tests for Binomial Distribution

If X has a $\text{BIN}(n, p)$ distribution, then for n large, one can use either the Poisson approximation theorem or the central limit theorem to obtain testing procedures for hypotheses about p . Thus, if n is large, but np is small, then asymptotically $X \sim \text{POI}(np)$. The hypothesis $H_0 : p = p_0$ corresponds to $H_0 : \lambda = np_0$ in the Poisson distribution. We can use tests for this distribution.

If neither np nor $n(1-p)$ is small, then we have $X/n \sim N(p, p(1-p)/n)$. To test the hypothesis $H_0 : p = p_0$ (against a one- or two-sided alternative), we use the fact that under H_0 , the distribution of the random variable

$$Z = \frac{X/n - p_0}{\sqrt{p_0(1-p_0)/n}}$$

is asymptotically standard normal.

For two sample situations, let $X \sim \text{BIN}(n_1, p_1)$, $Y \sim \text{BIN}(n_2, p_2)$, where n_1 and n_2 are both large, and assume that we can use normal approximation. Then we have

$$\frac{X}{n_1} - \frac{Y}{n_2} \sim N\left(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right).$$

Thus under the null hypothesis $H_0 : p_1 = p_2$, we have, letting $p_1 = p_2 = p$,

$$\frac{X/n_1 - Y/n_2}{\sqrt{p(1-p)(1/n_1 + 1/n_2)}} \sim N(0, 1).$$

This statistic still involves the nuisance parameter p . However, under H_0 the MLE of p is $(X + Y)/(n_1 + n_2)$, so we can use the test statistic

$$Z = \frac{\frac{X}{n_1} - \frac{Y}{n_2}}{\sqrt{\frac{X+Y}{n_1+n_2} \left(1 - \frac{X+Y}{n_1+n_2}\right) \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}},$$

which is asymptotically standard normal. Whether we use the one- or two-sided test depends on the alternative hypothesis.

■ EXAMPLE 12.36

In primary elections, 28% of the Republicans in New Hampshire voted for candidate A. A poll of 180 Republicans in Iowa show that 41 of them will vote for candidate A. Does this result indicate that the Republican support of candidate A is lower in Iowa than in New Hampshire?

SOLUTION. We have here $p_0 = 0.28$, the known level of support for candidate A in New Hampshire, and we want to test the hypothesis $H_0 : p \geq 0.28$ against the alternative $H_1 : p < 0.28$, where p is the fraction of Republican voters who support candidate A in Iowa. The observed value of test statistic is

$$Z = \frac{41/180 - 0.28}{\sqrt{(0.28 \times 0.72)/180}} = -1.58,$$

which corresponds to the p -value 0.059.

■ EXAMPLE 12.37

Continuing Example 12.36, assume that 110 of the 180 persons polled were women, and 20 of them (as well as 21 of 70 men polled) said they would vote for the candidate A . Does this result indicate that the support for candidate A in Iowa is higher among men than among women?

SOLUTION. We now want to test the hypothesis $p_W = p_M$ against the alternative $p_W < p_M$. The value of the test statistic is

$$z = \frac{\frac{21}{70} - \frac{20}{110}}{\sqrt{\frac{41}{180} \left(1 - \frac{41}{180}\right) \left(\frac{1}{110} + \frac{1}{70}\right)}} = 1.84.$$

The p -value here is about 0.033, and this may serve as an indication that indeed, the support for A among Republican women is lower than among Republican men.

12.10 MONTE CARLO, BOOTSTRAP, AND PERMUTATION TESTS

The tests introduced so far were based either on the known distribution of a test statistic or on its asymptotic form applicable when the sample size is large enough. There are, however, problems in which the distribution of the statistic is either unknown or difficult to handle theoretically, or the sample is not large enough to use the asymptotic properties of a test statistic. In all such cases, one could apply powerful computer-intensive statistical procedures that recently became readily available with advances of computer technology.

■ EXAMPLE 12.38

Assume that $X_1, \dots, X_n$ is a random sample from the Laplace distribution with a density $f(x; \mu, \lambda) = (\lambda/2) \exp\{-\lambda|x - \mu|\}$, and λ is known. If we want to test the hypothesis $H_0 : \mu = 0$ against the alternative $H_1 : \mu > 0$ at the significance level α , then for the large sample size, we could use the central limit theorem and reject H_0 if $\bar{x} > \sqrt{2}z_\alpha/\lambda$ (recall that $\text{Var}(X) = 2/\lambda^2$). However, if the sample size is rather small (e.g., $n = 10$), the normal approximation of the distribution of $\bar{X}$ may not be appropriate, and other methods should be used instead.

Monte Carlo Tests

In Monte Carlo tests, the critical values for given levels of significance, or conversely, the probabilities for specified thresholds, are estimated from generated samples based on the fact that relative frequencies can be used to estimate probabilities.

■ EXAMPLE 12.39

Continuing Example 12.38, if we are testing $H_0 : \mu = 0$ against $H_1 : \mu > 0$ at the significance level $\alpha = 0.1$, then we need to find a critical value v_α such that $P(\bar{X} \geq v_\alpha | \mu = 0) = \alpha$. For that we generate many (at least several thousand) samples of size $n = 10$ from the distribution with density $f(x; \mu = 0, \lambda) = (1/2)\lambda \exp\{-\lambda|x|\}$, and for each of them, we obtain its sample mean. Next, $\hat{v}_\alpha$ is obtained as the sample upper 0.1 quantile in the set of all means in generated samples.

In Examples 12.38 and 12.39, the sample size was small. In other situations, where the use of Monte Carlo tests is recommended, distributions of statistics are difficult to track theoretically (e.g., when population distributions are mixed or contaminated), or the estimators of parameters of interest are difficult to handle theoretically (e.g., sampling distribution of the coefficient of skewness, kurtosis, or trimmed mean).

Bootstrap Tests

Bootstrap procedures are designed for the situations where the population distributions are unknown, and the actual samples provide the only available information. Consequently, the reference distribution for the statistic of interest is obtained from bootstrap samples. One may notice that bootstrap sampling is actually a Monte Carlo sampling from the distribution obtained from all n sample values, with probabilities $1/n$ assigned to each of them (the generation of bootstrap samples was described in Section 11.7). The p -values of the test are obtained as appropriate relative frequencies from the bootstrap distribution of the statistic being used.

■ EXAMPLE 12.40

Let us consider a problem in which hypothesis $H_0 : \mu = \mu_0$ is tested against $H_1 : \mu > \mu_0$. The test statistic that can be used here is $\bar{X}$, and its sampling distribution is imitated (as explained in Section 11.7) by the bootstrap distribution. However, we need to keep in mind that the p -values must be obtained as probabilities for the distribution determined by the null hypothesis. Since we do not know what distribution the random sample represents (with $\mu = \mu_0$ or rather $\mu > \mu_0$), before generating bootstrap samples, we need to modify sample values by taking $w_i = x_i - \bar{x} + \mu_0$, where $\bar{x}$ is the mean obtained from the original sample. Then, the p -value will be obtained as

$$p\text{-value} = \frac{\text{number of } \bar{w}_i^* \text{'s exceeding } \bar{x}}{B},$$

where $\bar{w}_i^*$ is the average in the i th bootstrap sample.

We can also use bootstrap tests to compare parameters in two populations. In such a case, we usually have two independent random samples $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$. B bootstrap samples $X_1^*, \dots, X_n^*$ and $Y_1^*, \dots, Y_m^*$ are then generated by sampling with replacement from the same set that consists of combined $m + n$ original observations.

Permutation Tests

Permutation tests are similar to bootstrap tests as they also use only actual sample(s) to obtain a reference distribution based on which a p -value is determined. However, the sampling process is now different.

Practically, all parametric tests have a corresponding permutation test that is based on the same test statistic as the parametric test. But the p -value is obtained from the sample-specific permutation distribution of that statistic rather than from the theoretical distribution using a parametric assumption.

We will focus here on a two-sample test that compares two population distributions; the H_0 states that both distributions (or actually their respective parameters) are equal. An important assumption behind a permutation test is that the observations are exchangeable under the null hypothesis. The respective samples $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_m)$ of size m and n can then be combined into one set of size $m + n$ which is then partitioned into two subsets of sizes m and n so that the value of the appropriate statistic can be obtained. This process is repeated for all possible $\binom{m+n}{m}$ partitions, eventually giving the reference distribution based on which a p -value will be determined as a measure of how “extreme” is the value obtained from the original samples.

■ EXAMPLE 12.41

Let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_m)$ be two particular samples selected randomly from distributions $f(x; \theta_0)$ and $f(y; \theta_1)$, respectively. We want to test hypothesis $H_0: \theta_0 = \theta_1$ against the alternative $H_1: \theta_0 < \theta_1$. Assume that the difference $\Delta = \theta_1 - \theta_0$ is estimated by some statistic $\hat{\Delta}$, and its large values provide evidence supporting the alternative hypothesis. To perform the permutation test, we first obtain $\hat{\Delta}_0 = \hat{\Delta}(\mathbf{x}, \mathbf{y})$. Then samples $\mathbf{x}$ and $\mathbf{y}$ are combined into a set $\mathbf{w} = (x_1, \dots, x_n, y_1, \dots, y_m)$, which is next partitioned $\binom{m+n}{m}$ times into two subsets of sizes m and n , respectively. For each partition, the value of statistic $\hat{\Delta}$ is obtained and the p -value is determined as

$$p\text{-value} = \frac{\text{number of values of } \hat{\Delta} \text{ equal to or exceeding } \hat{\Delta}_0}{\binom{m+n}{m}}.$$

If sample sizes m and n are such that the number of all possible partitions $\binom{m+n}{m}$ is very large, then a smaller number of partitions can be obtained in a random (rather than systematic) way. The procedure is then called a *randomization test*.

More information on procedures introduced in this section can be found, for example, in Good (2005).

PROBLEMS

- 12.10.1** A random sample 0.38, −0.49, 0.03, 0.21, 0.12, 0.14, −0.18, −0.34, 0.46, −0.01 was selected from a Laplace distribution with density $f(x, \mu) = \exp\{-2|x - \mu|\}$. Use Monte Carlo simulations to estimate the p -value for testing the hypothesis $H_0: \mu = 0$ against the alternative $H_1: \mu > 0$.
- 12.10.2** A random sample 1.138, 1.103, 3.007, 1.307, 1.885, 1.153 was obtained from a distribution with density $f(x, \theta) = 2\theta^2 x^{-3}$ for $x \geq \theta$ and 0 otherwise. Find the MLE of θ , and use it to test $H_0: \theta = 1$ against $H_1: \theta > 1$. Find the p -value and compare it with the p -value based on 2000 Monte Carlo generations.
- 12.10.3** Assume that the hypothesis $H_0: \theta = 0$ is to be tested against $H_1: \theta > 0$, where θ is a parameter in the $U[\theta, \theta + 1]$ distribution. (i) Derive the test of size $\alpha = 0.05$, based on one observation only, and obtain its power function for $\theta = 0.1, 0.2, \dots, 0.9$. (ii) Use Monte Carlo simulations to obtain a test with a critical region $\bar{X}_3 > k$; that is, estimate k . Then again use Monte Carlo simulations to estimate the power function of this test at $\theta = 0.1, 0.2, \dots, 0.9$. Use the same size $\alpha = 0.05$. (iii) Compare the power functions of both tests.

12.10.4 Use the bootstrap test for testing $H_0 : \mu = 2$ against $H_1 : \mu > 2$, based on the random sample: 3.49, 2.21, 1.07, 3.04, 2.57, 2.43, 2.18, 1.10, 1.04, 1.92, 0.99, 3.13, 0.92, 2.72, 4.03.

12.10.5 (Shoshoni Rectangles) The following problem is taken from Larsen and Marx (1986). Since antiquity, societies have expressed esthetic preferences for rectangles having a certain width (w) to length (l) ratio. For instance, Plato wrote that rectangles formed of two halves of an equilateral triangle are especially pleasing (for such rectangles $w/l = 1/\sqrt{3}$). Another standard, adopted by the Greeks, is the golden rectangle. It is defined by the condition that the ratio of its width to length must be the same to that of the part remaining after cutting off a square with the side equal to its width (i.e., the shaded area in Figure 12.7 is similar to the whole rectangle). Thus, a golden rectangle must have $w/l = (l - w)/w$, which gives $w/l = \frac{1}{w/l} - 1$; hence $w/l = (\sqrt{5} - 1)/2 = 0.618$.

Both the Greeks and the Egyptians used golden rectangles in their architecture. Even today, the golden rectangle remains an architectural and artistic standard (e.g., items such as drivers' licenses, business cards or picture frames often have w/l ratios close to 0.618).

The data below show width-to-length ratios of beaded rectangles used by Shoshoni Indians to decorate their leather goods. Use the bootstrap test to check whether the golden rectangle can be considered an esthetic standard for the Shoshonis.

Width-to-length ratios for 20 rectangles found on Shoshoni handicraft:

0.693	0.749	0.654	0.670	0.662	0.672	0.615	0.606	0.690	0.628
0.668	0.611	0.606	0.609	0.601	0.553	0.570	0.844	0.576	0.933

12.10.6 A study of pollution was carried out in two lakes, A and B . The level of a specific pollutant was measured using a certain instrument, and the results for the lake A were 3.17, 4.22, 2.58, 4.01, and 3.79. In the lake B , the measurements (made with the same instrument) were 4.04, 4.32, and 4.12. Test the hypothesis that pollution levels are the same in both lakes, against the alternative that the pollution level in lake B is higher using: (i) The bootstrap test. (ii) The permutation test. (iii) Compare both results.

12.10.7 Two random samples 4.49, 7.68, 5.97, 0.97, 6.88, 6.07, 3.08, 4.02, 3.83, 6.35, and 4.59, 3.39, 3.79, 6.89, 5.07, 7.41, 0.44, 2.47, 4.80, 7.23 were obtained independently from distributions with the same mean. Perform a permutation test to test the

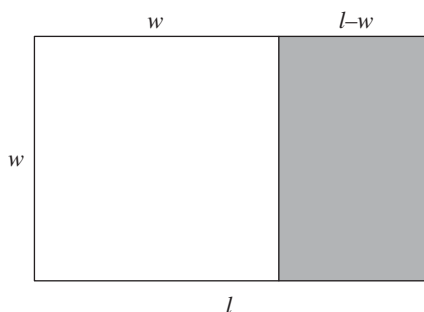


Figure 12.7 Golden rectangles.

hypothesis that the variability in both populations is the same against the alternative that it is larger in the second population. As a test statistic use: **(i)** The difference of sample ranges. **(ii)** The ratio of sample variances. **(iii)** Compare both results.

12.10.8 The following bivariate data show 16 independent observation of variables (X, Y).

X	0.95	1.30	2.55	1.60	2.70	2.77	2.85	1.37
Y	110	135	320	160	288	316	131	144
X	2.70	3.60	1.40	2.85	2.60	2.17	1.85	0.93
Y	279	340	191	301	280	302	258	125

Perform the bootstrap test to investigate if X and Y are associated. Use sample correlation as a test statistics.